

MASTER THESIS

Improving Radial Velocity Precision in the Near InfraRed with the NIRPS Spectrograph

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Abstract

With the current generation of high resolution spectrographs capable of reaching radial velocity (RV) precision of 1-m/s, the detection of an Earth-twin still remains a challenge. The RV signal induced on the sun by the Earth is 10-cm/s, a difficult detection limit to reach. However, for smaller, less massive stars like M-dwarfs, this signal is bigger and can be of the order of 1-m/s, while also positioning the habitable zones closer to the stars due to their low levels of insolation. In order to maximize the flux received from these M-dwarfs, the Near Infra-Red wavelength regime is ideal as this is where the black-body peak of these stars is. The Near Infra-Red Planet Searcher is a state-of-the-art echelle spectrograph operating in the wavelength range of 0.98 - 1.9- μ m capable of m/s RV precision with the goal to find earth-like planets in the habitable zone around M-type stars. It operates in conjunction with the HARPS spectrograph, together providing a full view of the Visible and Near Infra-Red high resolution spectrum. Several systematics, not significant in the visible, strongly impact the NIR spectrum and the derived RVs with the primary one being the Earth's atmospheric absorption and emission. The goal of this master project is to mitigate the impact of atmospheric emission by implementing a correction algorithm in the official ESPRESSO-NIRPS DRS, and the atmospheric absorption by modifying the weights in the Cross Correlation Function (CCF) masks according to the strength of the absorption features. The work done here would not only benefit the NIRPS collaboration but also the wider community as the data reduction pipeline is public.

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Chapter 1

Introduction

1.1 What is an Exoplanet?

'Exoplanet' is a portmanteau of the two words 'extra-solar' and 'planet', referring to a planet that orbits a star other than the Sun. The very first planet to ever be discovered orbiting a solar-type star other than the sun, 51 Pegasi b (Mayor and Queloz, 1995) was detected in 1994 and has since changed our perspective of the universe as a whole. Not only are there other solar systems out there, but their architecture can be vastly different from our own. Naturally, the next question that arose was: could there be another Earth, one that can support sentient life? To find an answer to this and many other queries related to the detection, characterization, formation and dynamics of planetary systems the field of 'exoplanetology' was developed. Throughout the years, several techniques have emerged that have successfully detected exoplanets, each having their strengths and weaknesses, and state-of-the-art instruments to improve the precision of the measurements. Of all the techniques, the one most relevant to this study is the Radial Velocity method that requires the use of an instrument known as a 'spectrograph' that splits the incoming light from a star as a function of wavelength to obtain the full spectrum for a particular wavelength regime.

1.2 Radial Velocity Method of Exoplanet Detection

51 Peg b was first discovered using this method of exoplanet detection and Figure 1.1 shows its working. If a star has a companion, the physics states that they would both orbit their common center of mass. The position of this center of mass depends on the separation between the two objects and their masses. For reference, Jupiter being the most massive object orbiting the Sun governs the Solar System's relative center of mass (also known as the Barycenter) which is located just outside the Sun's surface. Equation 1.2.1 gives the relation to calculate the minimum mass of the planet companion ($M_p \sin(i)$) using the measure velocity semi-amplitude (K), orbital period (P), planet mass (M_p), stellar mass (M_*), orbital inclination (i) and eccentricity of the orbit (e).

$$K = \left(\frac{2\pi G}{P} \right)^{1/3} \frac{M_p \sin i}{(M_* + M_p)^{2/3}} \frac{1}{(1 - e^2)^{1/2}} \quad (1.2.1)$$

If an instrument observes a star's spectrum over a wavelength range, one can compare the spectral lines and their wavelengths to the standard wavelengths where these lines should appear. If the star has a radial velocity component along our line of sight, due to the Doppler Effect, lines may be Redshifted (moving away) or Blueshifted (moving towards). To properly evaluate this shift in wavelength, the aforementioned spectrograph is used. Current state-of-the-art spectrographs are able to differentiate Doppler shifts to the precision of 1-m/s using an echelle grating as a disperser. Some high precision spectrographs currently operational are Echelle SPectrograph for Rocky Exoplanets and Stable Spectroscopic Observations (ESPRESSO, [Pepe et al. \(2021\)](#)), EXtreme PREcision Spectrometer (EXPRES, [Jurgenson et al. \(2016\)](#)), NEID ([Schwab et al. \(2016\)](#)), Keck Planet Finder (KPF, [Gibson et al. \(2016\)](#)), High Accuracy Radial velocity Planet Searcher (HARPS, [Pepe et al. \(2000\)](#)) and Near Infra-Red Planet Searcher (NIRPS, [Bouchy et al. \(2017\)](#))

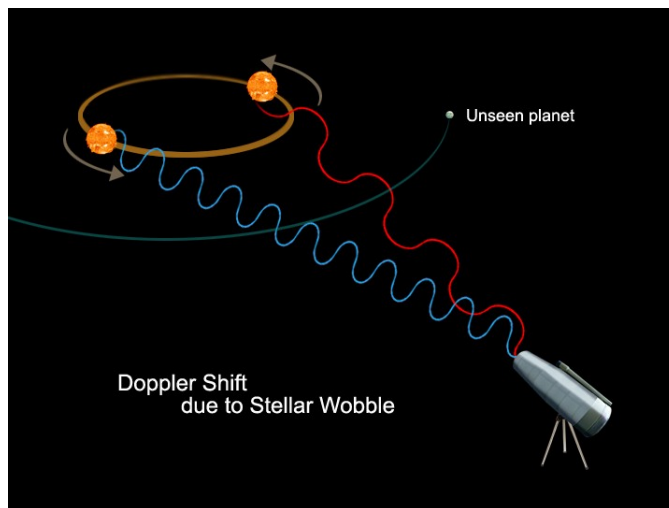


Figure 1.1: Radial Velocity method of planet detection ([NASA](#), nd)

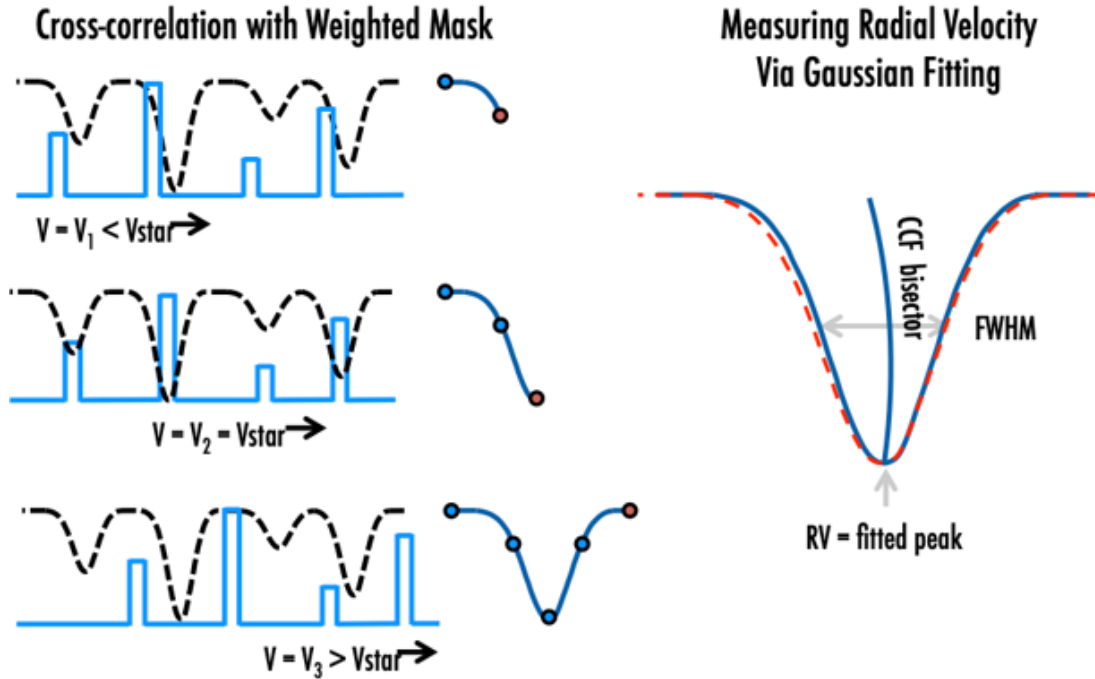
1.3 Radial Velocities through Cross Correlation Function (CCF)

Cross-Correlation is a technique that has been used for decades ([Baranne et al. \(1996\)](#), [Pepe et al. \(2002\)](#)) to obtain precise radial velocities (RV) for stars and hence detect planets around them. The two components needed to create a Cross-Correlation Function (CCF) are: spectrum of the target star and a 'mask' corresponding to the spectral type of the star. The CCF mask is a file that contains the location (as wavelengths) of all the strong absorption lines produced by a star of that spectral type as well as the 'weights' of these lines that give a measure of how deep the lines are relative to one another, the deeper the lines, the higher their RV content. These masks are usually constructed by repeatedly observing a bright target of a known spectral type, called the template star, over a long period of time to identify the spectral lines and their weight.

The exact process of computing a CCF can be inferred through Figure 1.3b. The CCF mask is 'cross-correlated' with the stellar spectrum in a way that the mask moves across a specified RV

1.4. NEAR INFRARED PLANET SEARCHER (NIRPS) SPECTROGRAPH

range and whenever it crosses over a spectral line that matches its own, a value of the 'contrast' is recorded according to the depth of the line, and everywhere there is no overlap, the level of the continuum is recorded. In the end we get a downward peak that acts as an 'average stellar absorption line'. By fitting an inverted gaussian to this CCF, one obtains the Full Width Half Maximum (FWHM) and most importantly the barycenter. This barycenter gives the value of the RV of the star at that epoch. Another parameter known as the bisector can also be obtained through the CCF profile.



(a) Construction of a CCF used for radial velocity computation (NEID, 2023)

Figure 1.2: Cross Correlation Function used to obtain precise radial velocities.

1.4 Near InfraRed Planet Searcher (NIRPS) Spectrograph

With the promise high resolution spectrographs have shown towards the detection and characterizing rocky exoplanets, the hunt for an Earth-twin continues strong. The radial velocity induced on the Sun due to the Earth is of the order of magnitude of 10-cm/s, still a challenging limit to reach due to several factors, one of which is stellar activity signals being one to two orders of magnitude larger than planetary signals at that level and the difficulties in disentangling the two. However, if one was to look at smaller, less massive stars like M-dwarfs, the amplitude of an Earth-size planetary signal increases (for the same distance from the host star). Another advantage of M-dwarfs is that their habitable zones are much closer to the star, due to their lower insolation, compared to that of a Sun-like star and since the radial velocity technique of planet detection favors close-in planets, the

chances of detecting said planet in the habitable zone is higher. Most new spectrographs operate in the visible wavelength range (350-nm to 800-nm) due to its many benefits, but M-dwarfs have their black-body peak in the infrared, so these spectrographs are unable to extract the maximum light possible.

The Near Infra-Red Planet Searcher (NIRPS) is a new state-of-the-art, fiber-fed, echelle spectrograph installed on the ESO 3.6-m telescope at La Silla Observatory, Chile. It operates in the near infrared (NIR) wavelength range of 0.98 - 1.9 microns, covering the Y, J and H bands. The distinct advantage of NIRPS is that it works simultaneously with the HARPS spectrograph, also mounted on the 3.6-m telescope. While HARPS works in the visible (VIS) wavelength band, when coupled with NIRPS, a simultaneous VIS and NIR spectrum analysis of the target star can be done. There are two optical fibers in the instrument, Fiber A which is pointed at the target (usually a star) and Fiber B, the reference fiber, that usually observes the sky background due to the ultra stability of NIRPS (or Fabry-Perot for concurrent reference) simultaneously with Fiber A. The two fibers are separated on the sky by a distance of 37-arcseconds. NIRPS offers a two instrument mode functionality, High Efficiency (HE) with a spectral resolution of $\sim 75,000$ using a 0.9-arcsecond fiber to observe a target star and a 0.4-arcsecond fiber to record the sky background, and High Accuracy (HA) offering a spectral resolution of $\sim 84,000$ with both the target and sky fibers being 0.4-arcsecond. HE mode allows for more light to enter the detector, using an image slicer to fit the 0.9-arcsecond fiber, at the cost of some spectral resolution and is primarily used for fainter targets and degraded seeing conditions. Due to lower modal noise in HE mode, it is almost always used for observations. The instrument saw its first light in May 2022 and the official observing period started in April 2023.

The NIRPS Consortium Science Team has an ambitious Guaranteed-Time-Observing (GTO) of 725 nights over 5 years, totalling 40% of the 3.6-m telescope time. It is jointly led by University of Geneva (Prof. François Bouchy) and University of Montreal (Prof. René Doyon) with other partners all over the world. The GTO time will be split between three Work Packages (WP) and Other Science (OS) programs. WP1 focuses on the blind search of planets around well known bright stars using the RV method. WP2 is the TESS follow-up program where transiting TESS Objects of Interest (TOIs) are observed with HARPS/NIRPS to confirm exoplanet detections. Finally, WP3 revolves around atmospheric detection and characterization of exoplanets.

NIRPS is unique in the sense that there are two viable Data Reduction Softwares (DRS), both capable of processing and reducing the raw spectrum into science-ready 2D and 1D spectra. While distinct in functionality, the existence of two DRS pipelines enables us to optimize the performance of both simultaneously, identify their individual strengths and weaknesses, and especially compare the reduced products and correct for any discrepancies or errors efficiently. The official DRS that functions at the telescope is a repurposed version of the ESPRESSO pipeline (hereby, ESPRESSO-NIRPS DRS), modified to work in the NIR regime. The current version of this pipeline is DRS-3.2.0 (released May 2024) which includes an updated version of the Earth's atmospheric emission and absorption correction, compared to the previous version (DRS-3.0.1). The other DRS pipeline is A PipelinE to Reduce Observations (APER0, [Cook et al. \(2022\)](#)), an initial re-coding of the HARPS pipeline, developed by the Canadian team as the official DRS for the SPectropolarimètre InfraROUge (SPIROU, [Artigau et al. \(2014b\)](#)) spectrograph, but now adapted to work on NIRPS data. APER0 is a more complex code that cannot process RV data at the telescope in a short amount of time (without human intervention), which is why it is used as a post-processing tool. The ESPRESSO-NIRPS DRS recognizes 71 spectral orders, while the APER0 pipeline recognizes 75, with the four extra orders being very weak in flux and not adding to the RV precision. It is

1.4. NEAR INFRARED PLANET SEARCHER (NIRPS) SPECTROGRAPH

important for an observer to obtain radial velocity measurements quickly from the observations just made, which is why at the telescope the RVs are calculated using the traditional Cross-Correlation Function (CCF) method, while Extreme Precision RV (EPRV) tools such as Line-By-Line (LBL) analysis are used in post-processing to go beyond the 1-m/s RV limit.

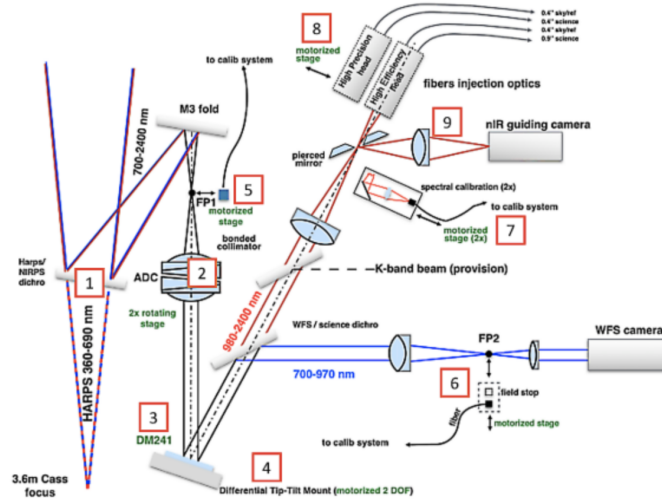
1.4.1 NIRPS Instrument Design

NIRPS utilizes a sophisticated optical design to achieve its remarkable spectral resolution (Cabral et al., 2022). The core element is an echelle grating, enabling the acquisition of high spectral resolution. Echelle spectrographs rely on the principle of diffraction. When light interacts with a regular pattern of microscopic structures, like the grooves on an echelle grating, it bends and separates into its component wavelengths, creating a spectrum. The key to the echelle spectrograph’s prowess lies in its unique grating. Unlike conventional gratings with closely spaced grooves, it has much wider spacing which allows it to exploit higher diffraction orders. The basic equation governing diffraction, known as the grating equation, relates the wavelength, the diffraction order, the spacing between grooves, and the angles of incidence and diffraction. By utilizing higher orders, the echelle achieves high resolution, meaning it can distinguish between very close wavelengths.

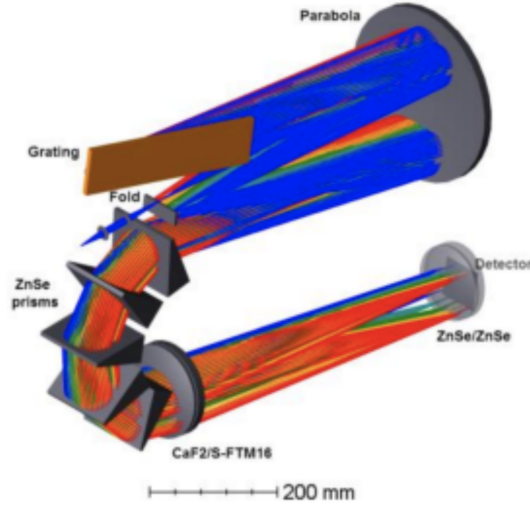
However, at higher orders there can be overlapping spectra. Blazing refers to the optimization of a grating’s groove profile to preferentially diffract light of a desired wavelength into a specific order. In the echelle, the grooves are typically designed with a specific angle on one face and a gentler slope on the other. This asymmetry directs most of the light’s intensity into specific higher orders, minimizing overlap with neighboring spectra and enhancing the overall signal-to-noise ratio. The echelle spectrograph employs a two-stage approach to separate the diffracted light. The first element, often a collimating mirror, transforms the incoming light into a parallel beam. The echelle grating then diffracts this collimated beam into its component wavelengths at high dispersion due to the higher diffraction orders. A second element, typically a prism or another grating, acts as a cross-disperser. This cross-disperser separates the overlapping higher orders spatially, allowing them to be captured by the detector, a NIR CCD camera.

NIRPS utilizes an echelle grating with a high number of grooves per millimeter, allowing it to achieve exceptionally high resolution. This high resolution also allows for analysis of the light from the star itself, revealing its chemical composition. In addition, NIRPS employs specialized infrared detectors for optimal performance in the near-infrared wavelength range. These detectors are highly sensitive to faint light, crucial for capturing the weak signals of exoplanets orbiting distant stars. This sensitivity is particularly important for detecting the subtle variations in a star’s radial velocity caused by an orbiting planet.

The Earth’s atmosphere acts as a barrier, distorting starlight through a phenomenon known as seeing. This effect would normally limit the achievable resolution of a ground-based instrument like NIRPS. However, it incorporates an innovative Adaptive Optics (AO) system that actively measures and corrects for these atmospheric distortions in real-time, effectively mitigating seeing and allowing the instrument to operate at a high-resolution even under the limitations of ground-based observations.



(a) Front-end design of NIRPS showcasing the path the light takes with respect to the different optical components including the AO system, guiding camera and the HARPS dichroic.



(b) Diffraction of the light entering NIRPS on the echelle grating, and its pathway to the detector.

Figure 1.3: Instrument design of NIRPS, aiming to produce high resolution NIR spectra and work alongside HARPS (Cabral et al., 2022).

1.5 Tackling Challenges of High Resolution Spectroscopy in NIR

Compared to VIS wavelength regime, NIR has a significant challenge in terms of ground-based spectrographs. This is because the Earth's atmosphere is not as transparent in the NIR as it is in the VIS. Several atmospheric molecular species such as H_2O , O_2 , CH_4 and CO_2 absorb the incoming light from a star, while the atmospheric OH radical and O_2 molecule can emit in the NIR, thus corrupting the stellar spectrum. However, recent techniques like [Artigau et al. \(2014a\)](#) and [Allart et al. \(2022\)](#) have found a way to automatically 'correct' the NIR spectrum from the atmospheric absorption *tellurics* by way of modelling these features. The other side of the coin is the atmospheric emission in the NIR cause by the OH radical and O_2 molecule present in the upper layers of the atmosphere. The emission features on the other hand went uncorrected in the ESPRESSO-NIRPS DRS as of the start of this project, while the absorption (hereby called telluric) lines were corrected using an algorithm designed by Roman Allart.

The main objective of this master project is to reduce the effect of both the telluric and emission features of the Earth's atmosphere on the radial velocity measurements obtained by NIRPS. The means by which this is achieved is different for both cases, and affect a different set of target stars. For the emission lines, we developed and implemented in the ESPRESSO-NIRPS DRS an algorithm that automatically corrects them from the science fiber A spectrum using the simultaneous fiber B sky observations and a master spectrum constructed of the locations of all the emission lines found in the NIRPS spectrum. While for the telluric features we looked into modifying on-the-fly the weights of certain lines of the CCF that were heavily contaminated by tellurics.

Chapter 2

Sky Emission Correction

2.1 Background on Atmospheric Correction in NIRPS

Prior to the start of this project, the ESPRESSO-NIRPS pipeline had a working version of the telluric correction for the absorption lines, but the emission features went uncorrected. The full extent of this atmospheric emission contamination on the NIRPS spectrum is displayed in Figure 2.1. These features (Figure 2.2) especially dominate the spectrum of faint stars, and since the majority of the TESS follow-up program (WP2) of the NIRPS GTO relies on detecting planets around faint M-dwarfs, it is imperative that there be a way to clean the spectra of these lines. Which is why the bulk of this work comprises of developing an algorithm to correct for the telluric emission lines in the fiber A of the NIRPS spectrum to work with the official ESPRESSO-NIRPS DRS pipeline. One option would be to ‘mask’ the artifacts, but because of their abundance in the NIR spectrum, correcting for them remains to be the only way to preserve the maximum amount of spectral information. The idea for the correction technique originated naturally through the evolution of the APERO pipeline’s reduction of SPIROU data, even though it was not mentioned in the initial APERO paper, and was relayed to the team in Observatoire de Genève by our colleagues in Université de Montréal via private communication. Despite these complications, coupled with engineering challenges, NIRPS is capable of delivering cutting-edge EPRV measurements and high resolution spectra to be used for ground-breaking exoplanet research.

2.2 Sources of Earth’s Atmospheric Emission

The night sky airglow (or emission spectrum) was being investigated as far back as the 1950s (Meinel, 1950; Bates and Nicolet, 1950; Anlauf et al., 1968). By obtaining, for the time, high resolution infrared spectra of the sky, a consensus was reached that it was primarily the rotational-vibrational modes of the OH radical that were responsible for the airglow. In the upper layers of the atmosphere, water vapour is bombarded by ultraviolet photons from the sun and dissociate into hydrogen and hydroxyl ions. These lone hydrogen atoms then collide with ozone molecules, and after several intermediate steps, produce hydroxyl ions following the chemical equation 2.2.1 where ν refers to the vibrational modes. These vibrational modes get excited during the creation of the OH , thus resulting in infrared emission. In addition to that, molecular oxygen, produced in

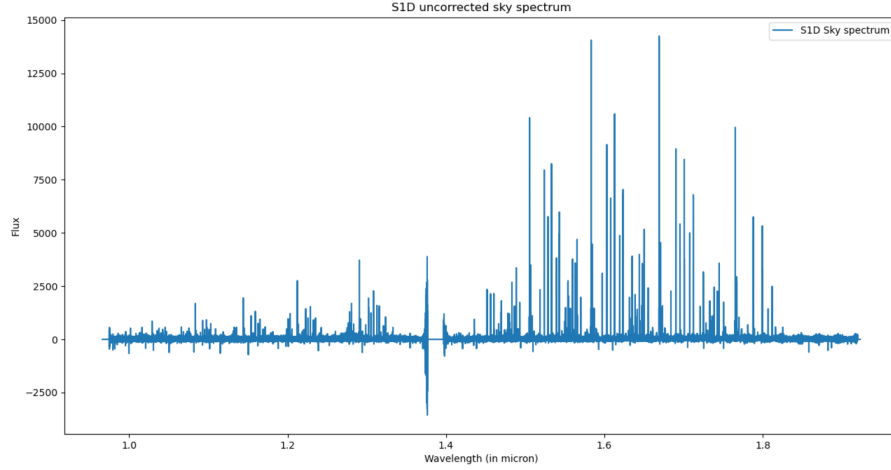


Figure 2.1: Uncorrected spectrum of the night sky across the full wavelength regime of NIRPS. Ideally this would be flat, but the atmospheric emission (upwards) and absorption (downwards) creates features that need to be corrected so as to not contaminate the spectra of science relevant stars.

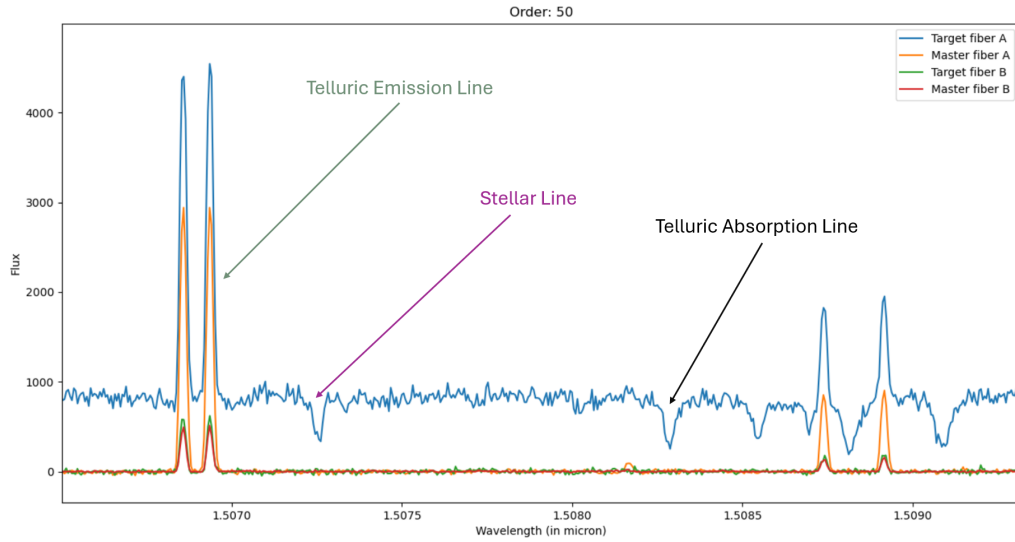


Figure 2.2: NIR spectral features seen in deblazed HE mode NIRPS spectral order 50 of TOI-5511. Target fiber A: Spectrum of TOI-5511, Master fiber A: Master spectrum of only sky emission lines, Target fiber B: Sky background observed simultaneously with TOI-5511, Master fiber B: Master spectrum of only sky emission lines

2.3. IDEA BEHIND ATMOSPHERIC EMISSION CORRECTION IN NIRPS

the same chemical reaction also emits in the small NIR band of $1.25\text{-}\mu\text{m}$ to $1.3\text{-}\mu\text{m}$ (Smith and Newnham, 1999).



2.3 Idea Behind Atmospheric Emission correction in NIRPS

NIRPS has the advantage of using two optical fibers in its observations. Fiber A observes the target, generally star, and Fiber B acts as the reference fiber and observes the sky, something only possible due to the ultra high stability of the instrument. Unfortunately the aperture of the two fibers is not the same in HE mode, thereby causing the spectra from Fiber A and Fiber B to have differing spectral resolution, so one cannot simply subtract the flux of the reference fiber from the science frames to obtain the 'sky subtracted' frames. To get around this issue with the resolution, we created a 'master' spectrum of the sky, for both fibers for both instrument modes, that contains the full library of nighttime emission spectrum. This master can then be scaled relative to the emission lines seen on the reference fiber during a science observation and subtracted from the Fiber A spectrum. Going a step further, this scaling and subtraction can be localized to a small wavelength region where an emission line is to prevent introducing additional noise to the entire spectrum. This technique heavily relies on two assumptions to hold true: the conditions of the atmosphere don't change over the $37''$ distance that separates the two fibers on the sky and the relative throughput (flux) between Fiber A and Fiber B does not evolve drastically with time. For the remainder of this work, the following terms will be used to differentiate the different spectra used to correct for the emission spectrum in NIRPS: Target A - Spectrum of the target (usually a star) observed via Fiber A, Target B - Nighttime spectrum of the sky simultaneously observed with the target via the reference Fiber B, Master A - High SNR master spectrum of the night sky containing the emission spectrum recorded on Fiber A, Master B - High SNR master spectrum of the night sky containing the emission spectrum recorded on Fiber B.

2.4 Construction of Master Sky Spectrum

Every few months, nighttime Sky-Sky (sky on both fibers) exposures are done with NIRPS for calibration purposes. These correspond to 6 x 10-minute exposures (60 minutes in total) on each night. So far we have a total of 30 Sky-Sky frames for HE mode, across 5 different nights, and 24 for HA mode, from 4 different nights, recorded over several months (January 2023 - October 2023). The exact frames used can be found in Table A.4.

A Sky-Sky frame contains information about the emission lines from the Earth's atmosphere so they are ideal for correcting for these features in the science observation. Ideally, one could simply do a sky observation right before the science one and then subtract the sky contribution from the science frame. However, it is not feasible to get a sky-sky calibration done every night. Also, an individual sky-sky calibration frame does not have the required signal to noise ratio (SNR) to accurately correct for the emission line features. Additionally, the emission lines can vary in intensity depending on the atmospheric conditions during the night. To solve these issues, we created a high SNR 'master' spectrum of the sky that combines the individual sky-sky frames over several months. There are in total four master spectra currently used in the ESPRESSO-NIRPS DRS pipeline due to the two instrument modes and the two fibers not having the same resolution.

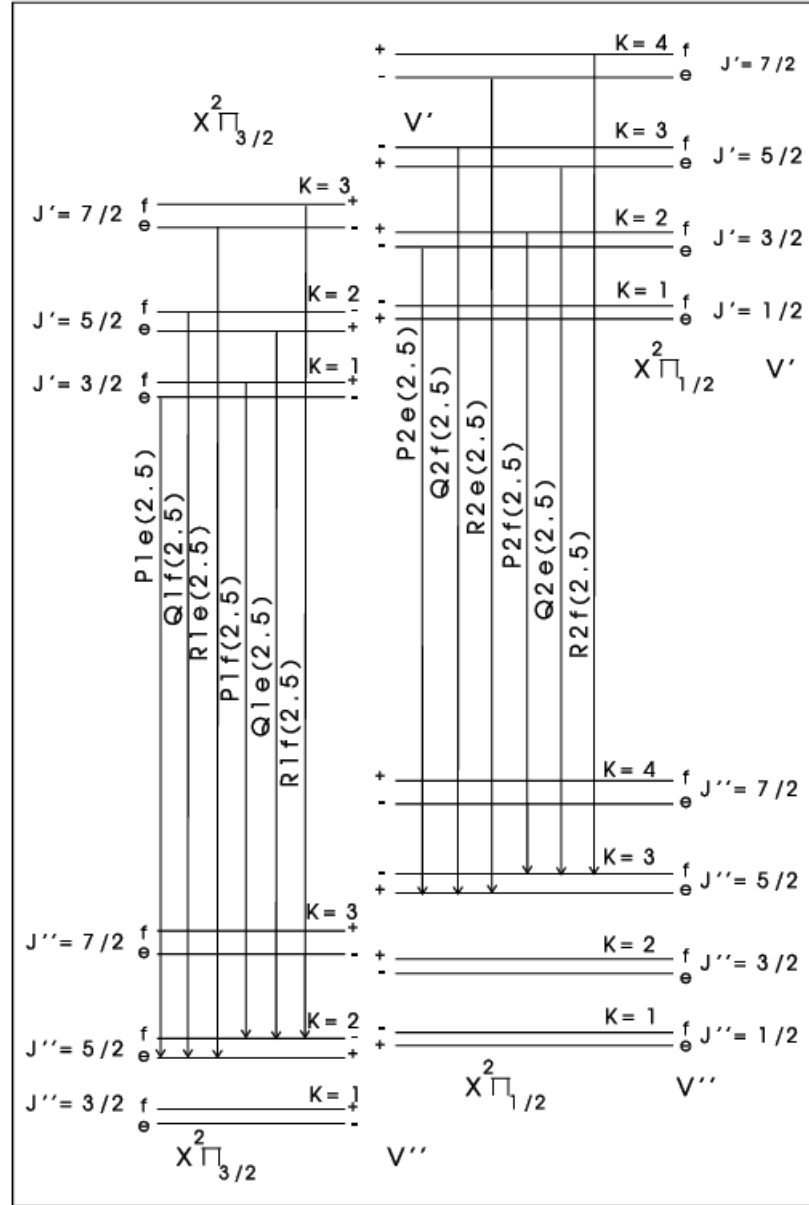
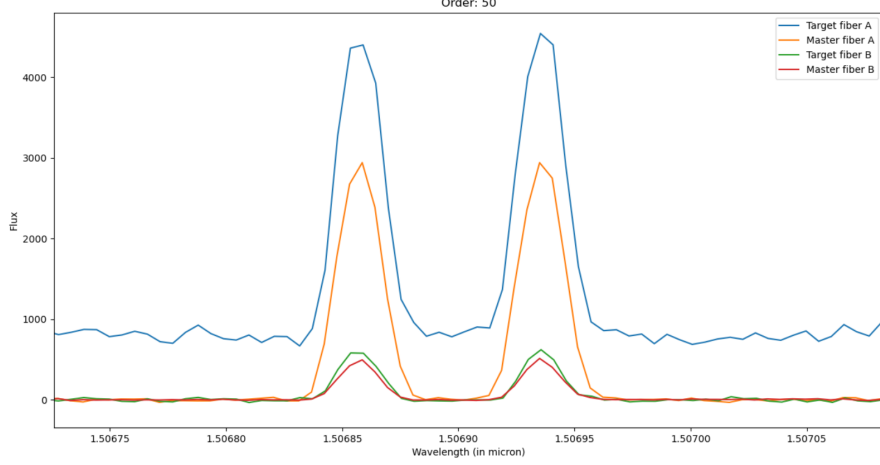
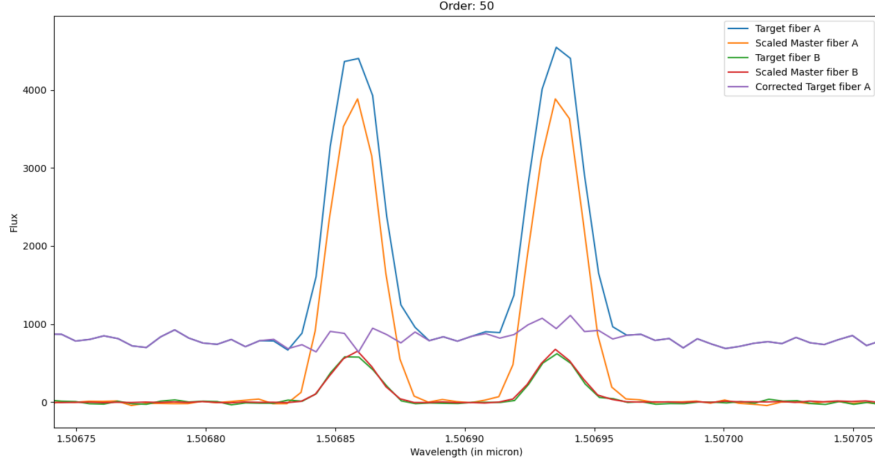


Figure 2.3: Vibrational transitions of OH responsible for NIR emission (Rousselot et al., 2000)

2.4. CONSTRUCTION OF MASTER SKY SPECTRUM



(a) Schematic of the spectrum of TRAPPIST-1 HE (Target) mode order 50 zoomed on an emission line doublet before the correction.



(b) Schematic of the spectrum of TRAPPIST-1 HE (Target) mode order 50 zoomed on an emission line doublet after the correction with both Master A and B called according to the ratio appropriate for this line (per-line or common).

Figure 2.4: Working of the emission correction on a doublet feature in the spectrum of TRAPPIST-1.

The master spectra therefore are as follows: HE mode Master A, HE mode Master B, HA mode Master A, HA mode master B. The process of creating these master spectra is identical for both instrument modes and fibers. The general process is highlighted below:

1. Collect all the sky-sky spectra of one instrument mode (e.g.: HE) and one fiber (A or B) that pass the overall quality control checks (QC =1, found in the header of the file).
2. These sky-sky frames are now stacked and interpolated on a common wavelength grid. This wavelength grid has a number of rows equal to the number of spectral orders since each order will have its own bounds for upper and lower wavelengths. These bounds are obtained by comparing the starting and ending wavelengths of all the sky-sky frames for that order, since the wavelength calibration solution is different for every frame (by $\sim 0.15 \text{ \AA}$ (approximately 1 pixel) over the course of several months of observations), such that the maximum of all the starting wavelengths acts as the lower bound of the common wavelength grid of that order and the minimum of the the ending wavelengths is the upper bound. This is done to prevent introducing any artifacts on the edges by the interpolation algorithm. After the wavelength grid for each order is defined, all sky-sky frames are interpolated using the scipy python package (Virtanen et al., 2020) and then for each pixel on this grid, the median flux of all the frames is calculated. This gives the 'master' spectrum.

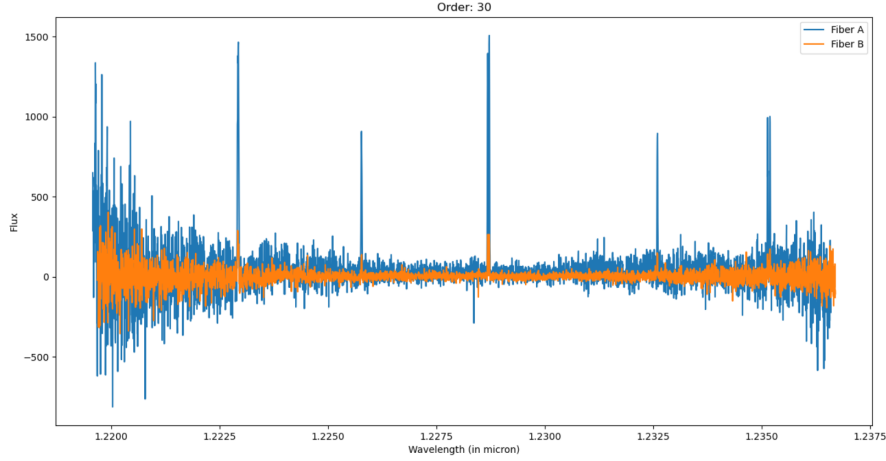
The obtained master spectrum has a higher SNR than all the individual sky-sky frames since the median would minimize the noise features of the individual spectra. In the beginning, this master spectrum was constructed using the deblazed sky-sky frames because the correction was originally intended to be performed on deblazed science telluric corrected spectra. However, during the official implementation several factors (Section 2.8) caused us to reevaluate that strategy and instead perform the emission correction before the telluric correction. Since the telluric correction required the blazed science spectra, and deblazing and re-blazing them before and after the emission correction was creating non insignificant discrepancies on the edges of each order, it was decided to keep the science frames blazed and use the blazed sky spectra as well for consistency.

2.5 Emission Line Selection

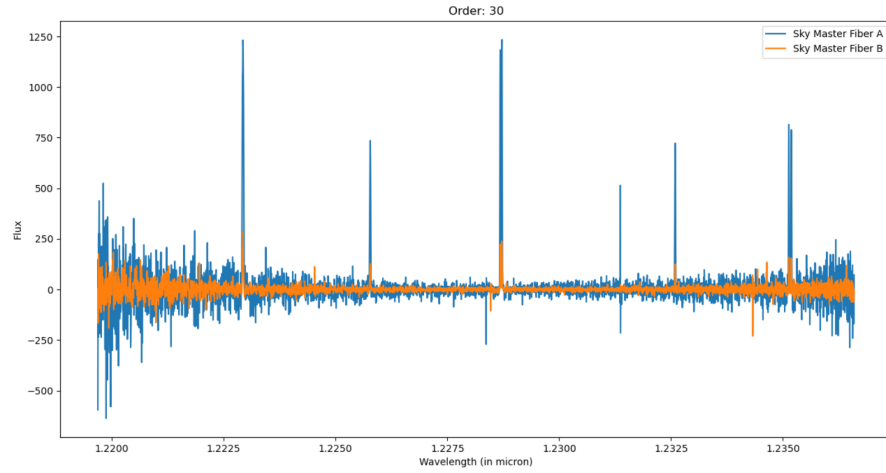
Due to the atmospheric contamination in the infrared, simply subtracting the spectrum on the sky fiber will introduce a large amount of noise into the data, especially for low SNR targets. Therefore, the correction for the emission lines needs to be limited to the position of the line itself, leaving the rest of the spectrum untouched. To facilitate this, identifying the correct lines is very important.

The automated process of locating the sky emission lines involves using a 'rolling' function from the 'Pandas' python package (Wes McKinney (2010), pandas development team (2020)) and compute the rolling median and Median Absolute Deviation (MAD). The advantage of using MAD over Standard Deviation (STD) is that MAD is less sensitive to the outliers, in our case the emission line features as shown in Figure B.2a. The window for the rolling function was picked to be 100 pixels. The logic behind the line selection process is that, the flux level of 'rolling median + n * rolling MAD' is significantly above the photon noise and the only points that lie above this limit are those from the emission lines. The value of 'n' was picked to maximize the balance between staying significantly above the noise level while also making sure the majority of the emission lines are included in the limit. For the NIRPS spectrum the value of 'n = 6' satisfied this condition the best.

2.5. EMISSION LINE SELECTION



(a) Low SNR deblazed and normalized single Sky-Sky frame of spectral order 30 in HE mode.



(b) High SNR deblazed and normalized Master Sky-Sky spectrum from spectral order 30 in HE mode. The noise level here is greatly reduced when compared to the figure above (especially at the edges) due to combining multiple sky-sky together.

Figure 2.5: Comparing the deblazed and normalized Sky-Sky spectra in HE mode as a single frame and as the final Master spectrum.

An artifact of using this method as an automated process of line detection is that in case of any detector systematics, readout errors or cosmic rays as in Figure 2.6a, where the flux values of one or two neighboring pixels can increase drastically, are also included in the tentative line list. Removing these features is critical so as to precisely correct only for the emission lines. Emission lines can exist in the spectrum in three forms: singlet, doublet and blended doublet, and as such if one was to model these lines with a gaussian profile using the lmfit python package (Newville et al., 2015), they would each have a different full width half-maximum (FWHM). The FWHM depends on the spectral resolution of the instrument, which for HE mode of NIRPS is approximately 4 pixels or 4-km/s (1-km/s = approximately 1 pixel for NIRPS) for singlets, and roughly double that for blended doublets. The relation between the spectral resolution the velocity shift corresponding to it is given in the following equation 2.5.1 where R is the spectral resolution of the instrument in a specific mode (HA/HE), δv is the velocity shift and c is the speed of light.

$$\frac{1}{R} = \frac{\delta v}{c} \quad (2.5.1)$$

If $R_{HE} \sim 75,000 \rightarrow \delta v \sim 4 \text{ km/s}$ and if $R_{HA} \sim 100,000 \rightarrow \delta v \sim 3 \text{ km/s}$. Plotting this distribution of the FWHM of all detected line candidates (see Figure 2.6b), the peak at 4-km/s corresponds to singlet lines. The other peak closer to 1 pixel instead represents all the cosmoics and detector systematics. Using this information, we filtered out all the line candidates with fewer than 2 pixels and created the definitive list of emission lines in the NIR spectrum of NIRPS. Again using the FWHM information, a 'line region' of +/- 1.5 FWHM was selected around each line in both fibers that acts as the part of the spectrum upon which the correction will be performed for that line.

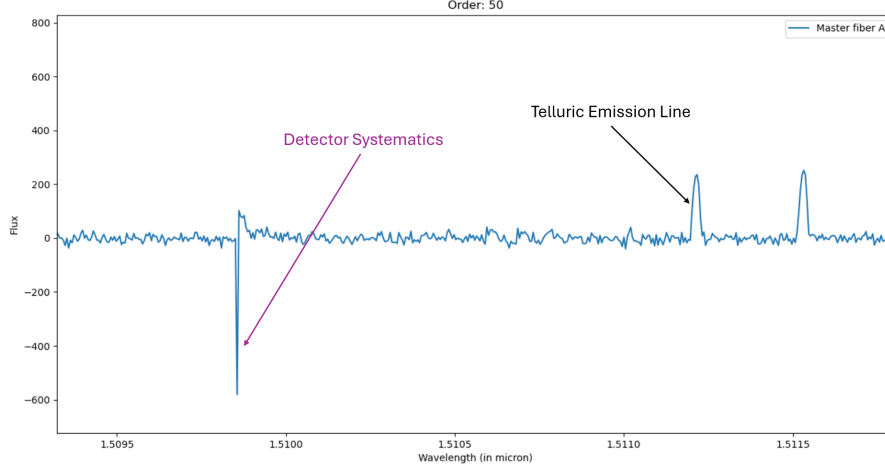
This process is repeated for both fibers of both instrument modes and then for each mode the line list is compared between fibers A and B. Since fiber A is larger (for HE) and allows more flux to enter, and has a higher efficiency (for both HA and HE), it would naturally detect more lines compared to fiber B. Often times the fainter ones would go undetected in B. There are only 293 line regions that are detected in both Master A and Master B. The final list can be found in Table A.2 with a total of 542 emission lines regions comprising of mainly wavelengths corresponding to OH and O2 molecules. In case of blended doublets and doublets too close to one another, both lines are considered a part of a single region to prevent the line regions from overlapping and causing complications in the correction code. Therefore, the actual number of individual lines detected is 657, where 534 are confirmed to be caused by OH radical, 61 by O₂ molecule, 40 blended lines that could belong to either or both species, and 22 lines from species that are neither OH or O₂. The identification of the species was done using the HITRAN (I.E. Gordon et al., 2022) catalogue.

2.6 Correction Method

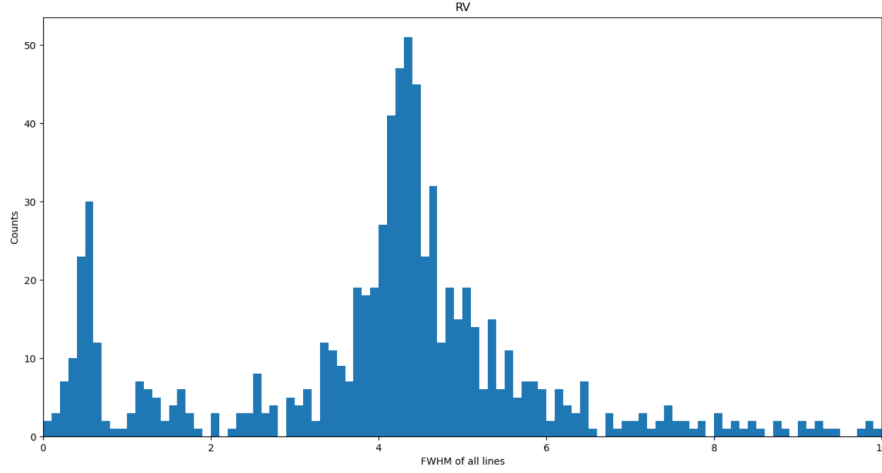
As mentioned previously, the atmospheric emission lines from a science frame are corrected by subtracting the master sky-sky spectrum from it after scaling it appropriately. The process of this subtraction is as follows:

1. A target frame is selected that contains the *S2D_BLAZE_A* and *S2D_BLAZE_B* files (full list of ESPRESSO-NIRPS DRS science products relevant for this project can be found in Table A.3). The fiber A file contains the spectrum of the target star, contaminated with

2.6. CORRECTION METHOD



(a) Showcasing detector systematics that might be falsely considered as emission line candidates by the automatic code.



(b) Distribution of Full Width Half Maximums of the Gaussian fits done to the emission line in candidates in HE mode master sky spectrum. The first peak is related to the detector systematics and the second one (at ~ 4 -km/s) corresponds to the singlet emission lines. All the higher values of the FWHM are for the blended lines.

Figure 2.6: Technicalities of the line selection process. It is important to distinguish the detector systematics from the real emission features and the FWHM is the method used in this work.

atmospheric emission features. The fiber B file on the other hand contains the nighttime sky spectrum with only the emission lines.

2. A rolling MAD calculation is done on both Target B and Master B. Then on each of them, a check is done to exclude any lines that are too faint. The limit of 6 MAD is used once again for consistency. Then the lines that passed the test for both Target B and Master B are compared and the common ones are selected.
3. Only for these common lines, the flux ratio between Target B and Master B is calculated for each of them by summing the net flux in the respective line region (called the Integrated Flux) and using the equation: $\text{ratio} = \text{flux target b} / \text{flux master b}$. A median of these ratios is then computed, henceforth called the 'Median Ratio' (also called the common ratio). This median ratio will be used to correct for the fainter lines in the science observation that are not seen on fiber B and only present in the fiber A spectrum. It should be noted preliminary testing was also done using a 'wave-range' ratio instead of the median ratio, but was discarded in favor of the median ratio. See Section 2.11.1 for detailed information.
4. Next, all the lines were sorted into two categories. Ones that would be corrected using their individual flux ratios (Per-Line ratios) and others that would be corrected using the median ratio. The lines not detected in both Target B and Master B are automatically added to the category of correction using the median ratio. An additional check is done on the per-line ratio to make sure they are not absurd, i.e., if the ratio is very close to zero, and also within the specified bounds: $0.33 * \text{common ratio} \leq \text{ratio} \leq 1.66 * \text{common ratio}$ (Figure B.1). If the remaining lines are in agreement with both of these conditions, they are corrected using the per-line ratio, otherwise the common ratio is used.
5. To do the actual line correction, a spectral order is first selected and then an emission line that is present in that order. The location of this line is then cross-matched with the previously constructed list of lines in the master and whether the common ratio or the per-line ratio is to be used for its correction.

To perform the correction using the common ratio or per-line ratio, the line region is taken from the line list and the closest starting and ending wavelengths are located on the Target A wavelength solution. Then the Master A line region is interpolated to match the wavelength grid of previously obtained line region in Target A. Now that the Master A and Target A spectrum share the same wavelength solution for that region, the flux on each pixel in the Master A line region is multiplied by the common ratio or per-line ratio, depending on the line, to scale it before subtracting this new flux from the Target A spectrum following equation 2.6.1.

$$\text{Target A Corrected} = \text{Target A} - \text{Ratio} * \text{Master A} \quad (2.6.1)$$

6. This process is repeated for every line in each order as well as every new frame because the per-line ratios and number of common lines can differ significantly depending on the atmospheric conditions of the night. The only static calibration files are: Master A and Master B spectra, the line regions and the integrated flux of the Master B lines.

Once this is completed for all the spectral orders of a frame, the telluric correction can be performed on it as the next step in the DRS. Since this process is equivalent to the 'sky subtraction'

done on ESPRESSO, for consistency, the emission corrected spectra are labeled as *SKYSUB* in all the DRS output files.

2.7 Error Propagation

Since the flux in the science spectrum is altered locally during the process of the emission line correction, it is important to propagate the error bars as well to reflect this change. Line regions with strong emission lines should have relatively larger error bars because the errors from the master spectrum need to be propagated into the emission corrected spectra.

2.7.1 Master Sky Spectrum Flux Error

Each frame that went into the creation of the Master Sky spectrum had the exact same exposure time, which implies that their SNR should be comparable within the limits of the sky variation. Equation 2.7.1 shows the calculation of the error for each pixel of the master spectrum (σ_{master}) with σ_{frame} being the error bar of that pixel for a given frame. The flux errors for each sky frame used to create the master are stored per-pixel as the 'ERRDATA' in the header of the DRS fits files. This is in theory not an exact propagation because of the individual frames being interpolated to match a common wavelength grid before the Master spectrum was created, but on NIRPS the distance between two pixels is roughly 1-km/s, which is small enough compared to the width of a typical emission line (4-km/s for HE mode) to carry over the error bars of the pixel before the interpolation without introducing any significant bias.

$$\sigma_{master} = \sqrt{\frac{1}{\sum \frac{1}{\sigma_{frame}^2}}} \quad (2.7.1)$$

2.7.2 Emission Line Corrected Spectrum Flux Error

Because the correction process is a simple subtraction of two spectra, locally, the net error of the final corrected spectrum would amount to the error of the two spectra added in quadrature. However, Master A is scaled according to a ratio, therefore so do the errors of Master A scale accordingly. The calculation of the errors of the corrected spectrum σ_{corr} is highlighted in equation 2.7.2, where R is the ratio by which Master A was scaled and σ_{target} is the error for each pixel of Target A obtained from the 'ERRDATA' header of the fits file of the spectrum.

$$\sigma_{corr} = \sqrt{\sigma_{target}^2 + (\sigma_{master} * R)^2} \quad (2.7.2)$$

2.8 Order of Implementation of Telluric and Emission Corrections

Cleaning the NIRPS spectrum of Earth's atmospheric contamination is a two-fold process. The emission and absorption features are corrected for separately, and done so using different methods. While in theory this should be treated as one 'step' in the DRS, in reality that is not possible, and as such the order of implementation of the two corrections matters. In APERO the telluric

correction precedes the emission correction, which is why during our implementation phase we only corrected the telluric corrected spectra. When it was time to adapt the emission correction into the DRS, it was agreed that the reduced *S2D.BLAZE* spectra would first go through emission correction and then telluric correction. There are several scientific and technical reasons why this was done, the Master Sky spectrum was constructed using telluric uncorrected sky observations, the emission correction is the equivalent of *SKY SUB* done in ESPRESSO which happens before telluric correction, and also occasionally the telluric correction fails to work so it is easier to implement the emission correction before it.

Comparing both scenarios (telluric correction first, then emission correction and vice versa) most of the time the spectra are identical, within error bars. However, in the locations where there are strong emission lines, the procedure where the emission correction was performed first corrects the line better (see Figure 2.7). Since the emission correction is applied locally, around the region of the detected emission line, it is possible that the discrepancy is only seen around the emission features as the telluric absorption correction is done on the entire spectrum and not around just the absorption features.

2.9 Quality Control for the Official Data Reduction Pipeline

The final step before implementing the emission line correction to the official ESPRESSO-NIRPS DRS was to set the quality control (QC) checks. These checks are important to ensure that anyone working with the science frames only uses them if the correction is performed correctly and to the desired precision. The QC flags used for the emission correction are: number of lines used to calculate the common ratio, upper and lower limits for the common ratio and the MAD limit of the per-line ratios.

There are many factors that come into play when trying to differentiate the emission features of one science frame from another, however, the goal is to define QC parameters that are not frame-specific. The most important criterion related to the emission correction that fits this category is the number of lines used to compute the common ratio *nb_lines*. It is obvious that the value of the common ratio and the MAD of the ratios (our QC parameters) are correlated with *nb_lines* (Figures 2.8 and 2.8a), and to set the correct QC values we had to remove this correlation.

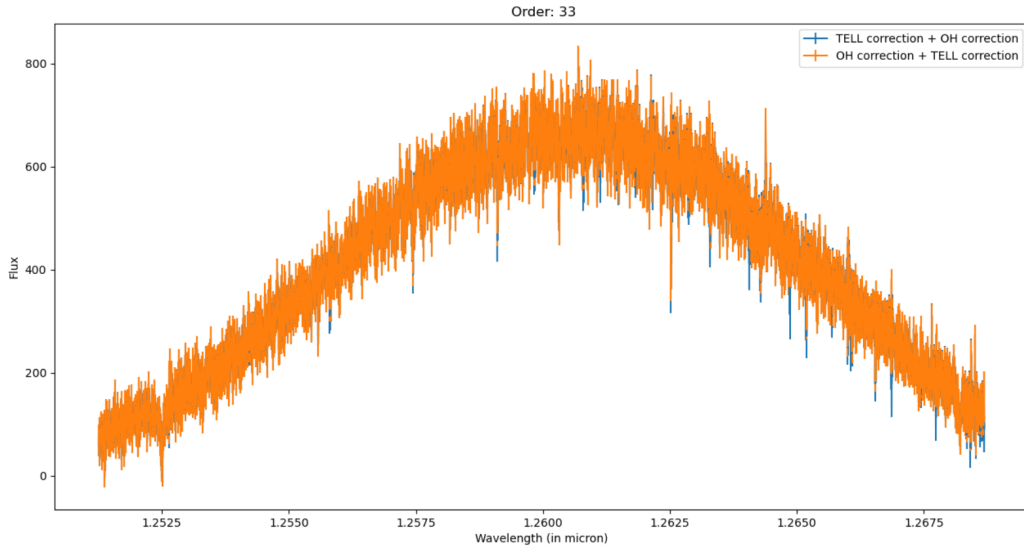
Since the common ratio compares the flux between the science frame fiber B and master fiber B, the most likely reason for the disparity between this flux level is the exposure time. The higher the exposure time, the more flux is recorded by the instrument, the more lines would pass the 6 MAD level to be used for the computation of the common ratio. The exposure time for all the sky-sky frames used for the master is the same, 601.9164 seconds with the telescope pointed close to the zenith, so we scaled the two QC parameters with this exposure time following the equations 2.9.1 and 2.9.2.

$$Normalized\ Common\ Ratio = Common\ Ratio * \frac{601.9164}{T_{exp}} \quad (2.9.1)$$

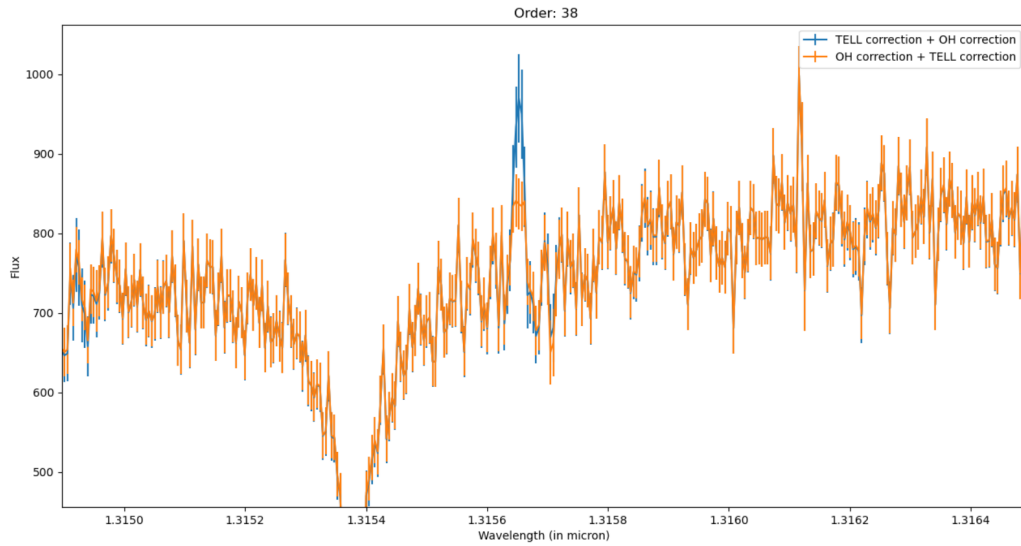
$$Normalized\ Ratio\ RMS = Ratio\ RMS * \frac{601.9164}{T_{exp}} \quad (2.9.2)$$

However, despite this, there was still some dependence on *nb_lines*, so another scaling factor was added to scale the QC parameters called the 'Airmass'. Airmass is a measure of the volume of air the light of a target star has to pass through, along the line of sight, to reach the telescope. The

2.9. QUALITY CONTROL FOR THE OFFICIAL DATA REDUCTION PIPELINE



(a) Spectral order 33 of a frame of Trappist-1 with the two orders of implementation of telluric and emission corrections (also called OH correction) giving an almost identical spectrum.



(b) Spectral order 38 of a frame of Trappist-1 with the two orders of implementation of telluric and emission corrections (also called OH correction) correcting the emission line at $1.3157\text{-}\mu\text{m}$ to different degrees.

Figure 2.7: Two spectral orders of TRAPPIST-1 comparing the order of implementation of the telluric and emission corrections.

higher the airmass of a science observation, the higher the level of atmospheric contamination, the stronger the emission lines. The parameters are therefore scaled using the revised equations 2.9.3 and 2.9.4.

$$\text{Normalized Common Ratio} = \text{Common Ratio} * \frac{601.9164}{T_{exp}} * \frac{1}{\text{Airmass}} \quad (2.9.3)$$

$$\text{Normalized Ratio RMS} = \text{Ratio RMS} * \frac{601.9164}{T_{exp}} * \frac{1}{\text{Airmass}} \quad (2.9.4)$$

With this, the correlation of the QC parameters with *nb_lines* is drastically reduced (Figure 2.8a). There is still a hint of a slope implying a further correlation, but for the purposes of setting the QC flags, it can be ignored for the time being. The following QC limits were finalized: minimum Normalized Common Ratio = 0.2, maximum Normalized Common Ratio = 3.5, maximum ratio RMS = 0.5, minimum number of lines used to compute common ratio = 10.

2.10 Validation of Emission Correction

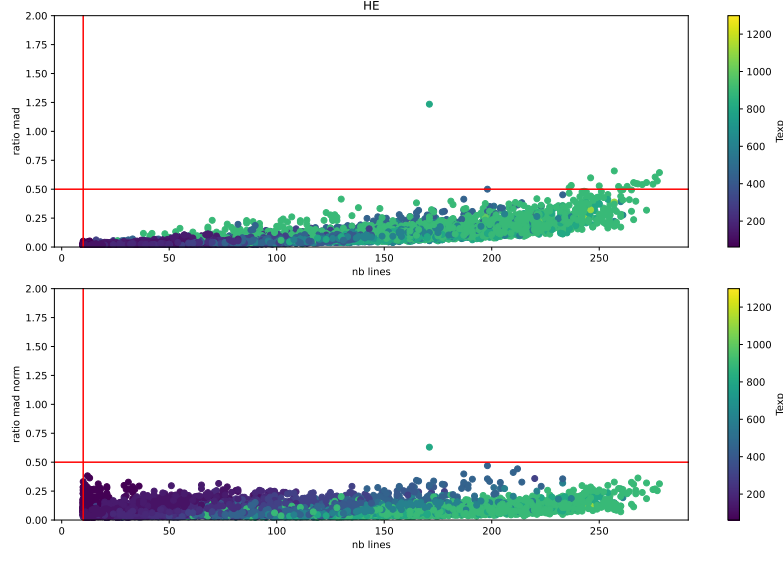
With the development of the emission line correction for the ESPRESSO-NIRPS DRS, it is important to test it across a variety of target stars in order to ensure that it is working as intended. The uncorrected emission lines heavily impact low SNR spectra of faint stars as the brighter stars have such high levels of measured flux that the atmospheric emission lines get blended in. Despite this, the correction algorithm should not introduce any artifacts into the spectra of bright targets. Additionally, the fiber size difference between the two instrument modes (HA and HE) may cause complications. The following targets were extensively tested for the verification: Proxima Centauri (Bright, HE and HA), TOI-3494 (Faint, HA), TOI-4666 (Faint, HE), TRAPPIST-1 (Faint, HE), TOI-756 (Faint, HE). The first version of the emission correction worked as a python script as a post processing to the DRS products. But with its integration in the DRS it is also key that both versions of the correction, despite being in different programming languages and having slightly nuanced implementation, agree in terms of the output.

2.10.1 Proxima (Bright, HA and HE mode)

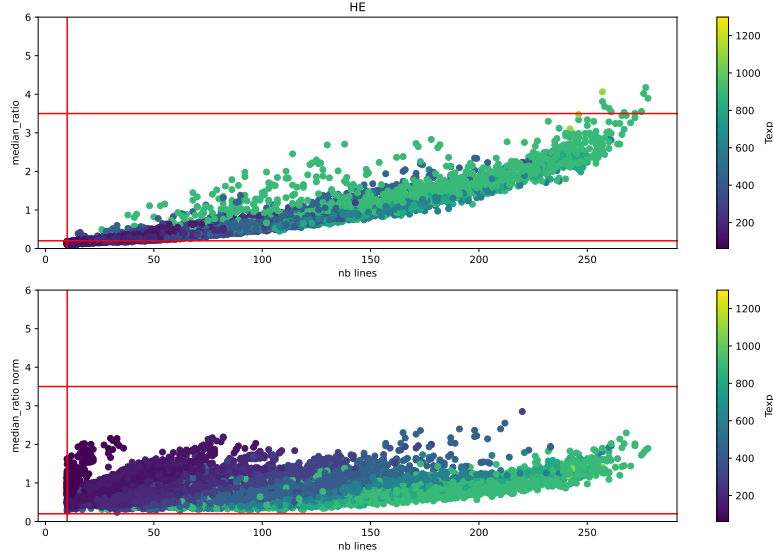
Proxima is one of the 'golden targets' that has been observed by NIRPS a considerable amount starting from the commissioning period. It is a bright M5.5 type star and has more than 300 observations in HE mode and 150 for HA mode, making it a prime candidate for judging the RV precision of NIRPS in a timeseries. Figure 2.9a showcases the differences between the emission corrected, emission uncorrected, and telluric corrected, spectra of Proxima in the spectral order 30. For such a bright star it is the telluric contamination that has a much bigger impact on the spectrum and creates inaccuracies in RV as opposed to emission lines, as it is only by zooming in that the effect of the emission correction is noticeable. Since the CCF is considered to be an average of all the stellar absorption lines, any change in its shape can be attributed to the telluric or emission correction algorithms where once again the effect of the emission correction is negligible ($\sim m/s$ level) as expected.

The next Figure 2.9b highlights the differences in the emission corrected spectrum from the preliminary python code (labeled 'Avidaan') and the official pipeline (labeled 'DRS') by plotting the flux ratio on spectral order 30 for HE mode. They are almost identical within the error margins

2.10. VALIDATION OF EMISSION CORRECTION

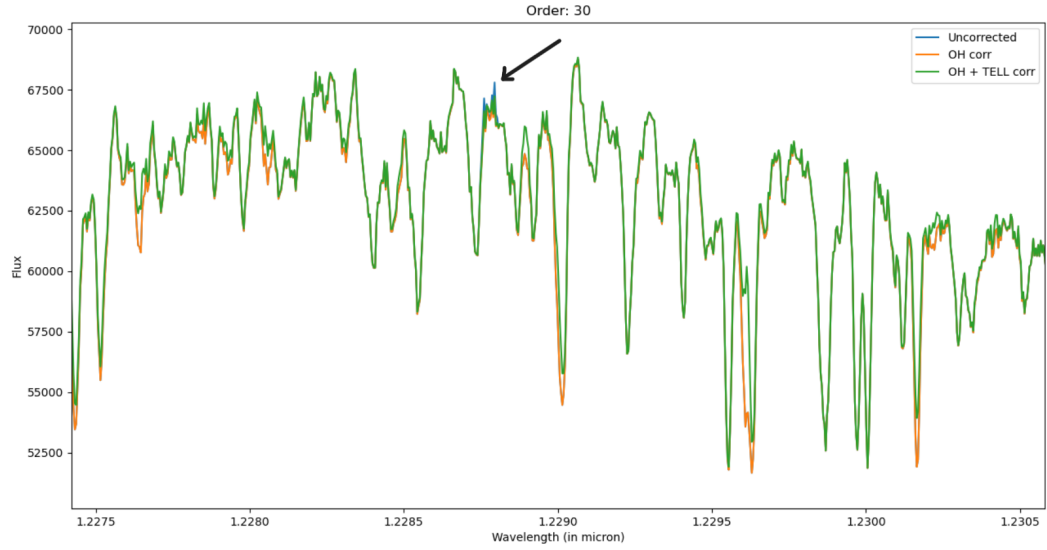


(a) Normalized MAD Ratio QC bounds (in red) on HE mode spectra. The goal of the normalized MAD is to de-correlate exposure time and number of lines used to compute the common ratio to make all the frames comparable for QC checks.

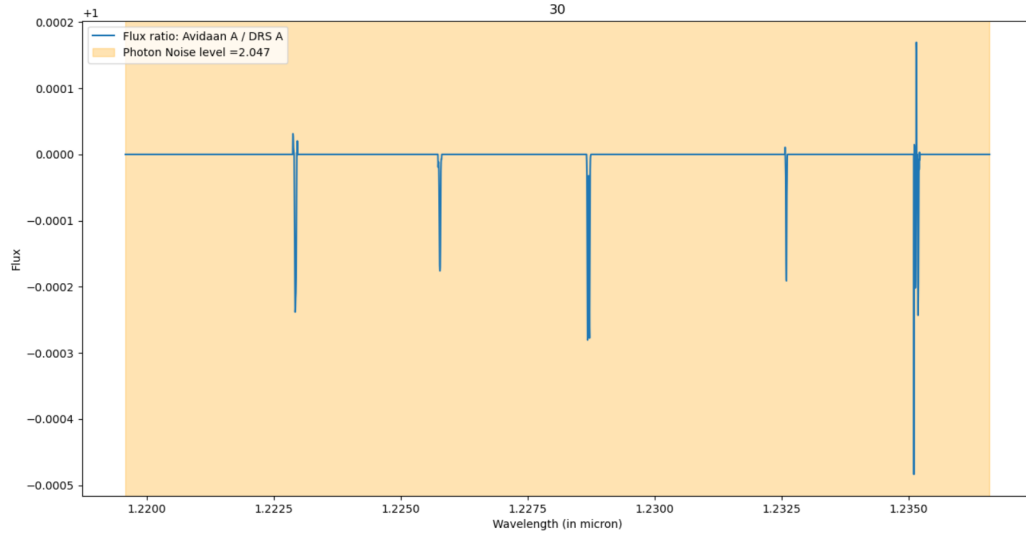


(b) Normalized Median Ratio QC bounds (in red) on HE mode spectra. The goal of the normalized median ratio is to de-correlate exposure time and number of lines used to compute the common ratio to make all the frames comparable for QC checks. It seems there is still some correlation, but it is reduced enough to define a tight QC threshold.

Figure 2.8: QC bounds on the Normalized MAD Ratio and Normalized Median ratio regarding the emission line correction. All data from 2022, 2023 and 2024 from NIRPS is represented on this figure.



(a) Different stages of reduction of Proxima spectral order 30 compared. Only after zooming in is the effect of emission correction (blue part) visible for such a bright target.



(b) Comparing the difference between the emission correction done by the python script and the DRS. The discrepancies ($\sim 10^{-4}$ units) are well under the photon noise levels (2.047 units).

Figure 2.9: Spectral order 30 of Proxima observed in HE mode.

of the flux, with the only differences localized to the regions where the emission correction was performed, and after calculating and comparing the CCFs (Figure B.5), the difference in RV comes out to be approximately 0.01-m/s while the RV error estimated by the photon noise level is about 2-m/s. The difference is well within the error bars and it can be concluded that for Proxima both the preliminary python script and the implemented DRS reduction are in agreement. The mask used to generate the CCFs from the ESPRESSO-NIRPS DRS was the M4 mask.

The advantage of having two different DRS to reduce NIRPS data is that they can be compared. Hence, Figure 2.10 over plots the final emission + telluric corrected spectra obtained from the ESPRESSO-NIRPS DRS and the APERO DRS for order 30. There is a 10 - 20% flux difference between the two due to some differences regarding the size of the window used to extract the flux from the detector. Ignoring that, the spectral features from both versions of the DRS appear to be in agreement. In the end, the DRS-3.2.0 timeseries of Proxima is improved on the m/s level from the DRS-3.0.1, which is to be expected for such a bright target (Figure 2.11).

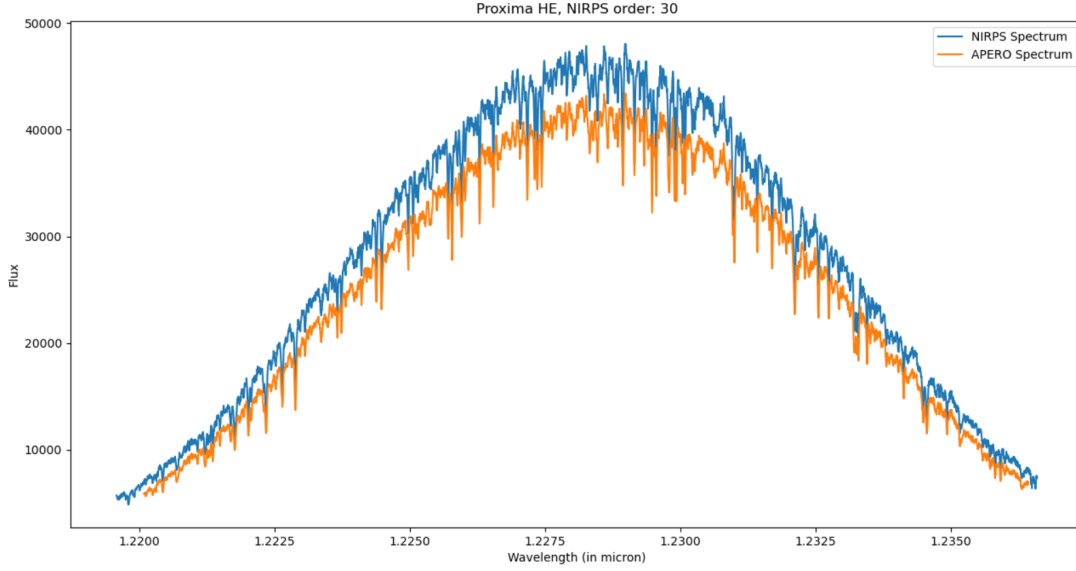


Figure 2.10: Comparing the emission and telluric corrected spectra of Proxima HE mode spectral order 30 produced by the ESPRESSO-NIRPS DRS and the APERO DRS.

2.10.2 TOI-3494 (Faint, HA mode)

As mentioned previously, the atmospheric emission features mainly contaminate the fainter stars, and TOI-3494 is one such M4 type star observed extensively using the HA mode of NIRPS. As seen in Figure 2.12a, the emission features are well corrected and the effect of the correction is easier to see in this case compared to Proxima. This correction also affects the CCF and RVs more significantly (Figure 2.12b), where before the correction the CCF was heavily biased by the presence of emission lines. This also reflected on the timeseries (Figure 2.13) where DRS-3.0.1

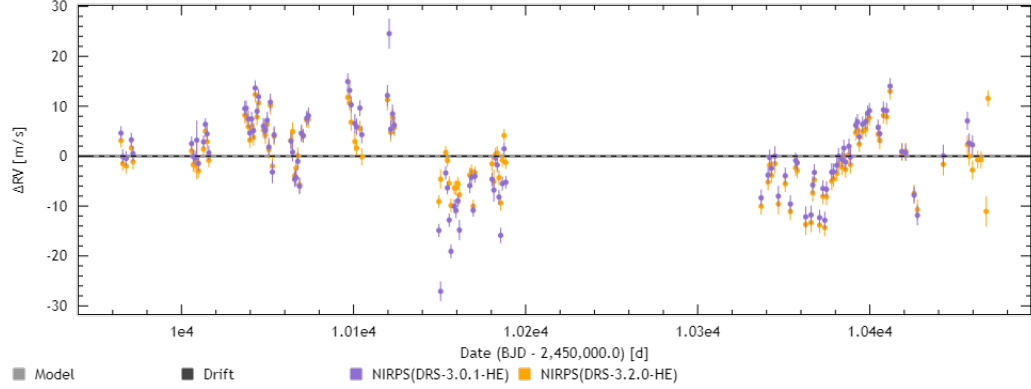


Figure 2.11: Proxima HE timeseries comparing the RVs from two versions of the DRS: DRS-3.2.0 (with emission correction) and DRS-3.0.1 (without emission correction). They are both in agreement, with DRS-3.2.0 offering a m/s level improvement.

shows huge levels of scatter due to the contaminated CCF while DRS-3.2.0 is in agreement with TESS ephemeris.

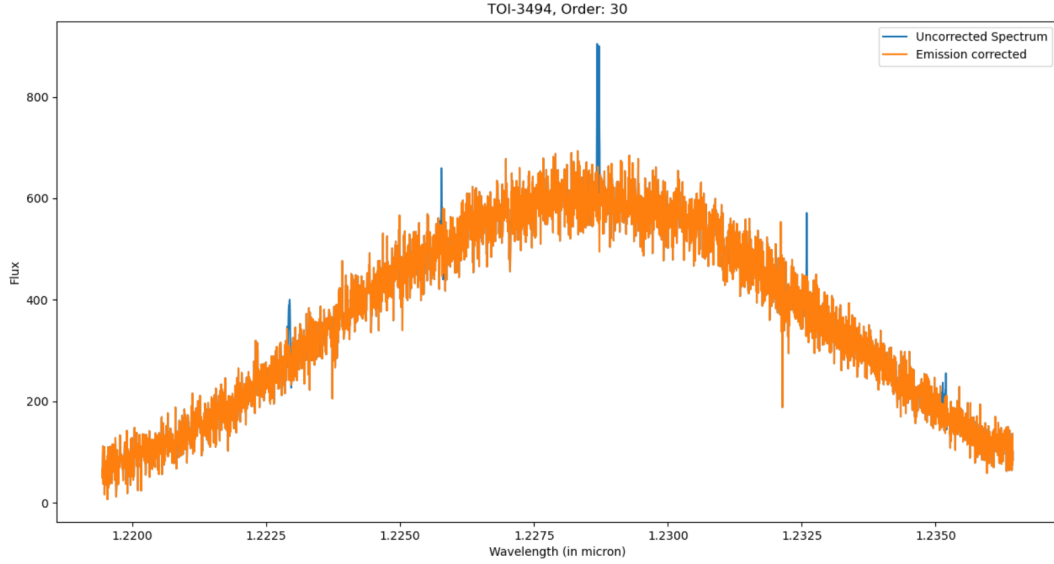
The python script and the DRS emission corrected spectra for the two spectral also give consistent results. The RV difference derived from the CCFs is ~ 0.5 -m/s while the photon noise RV error is ~ 20 -m/s, so well within the bounds to be considered consistent, both CCFs obtained using the NIRPS M4 mask. In order 30 at $1.229\text{-}\mu\text{m}$ the emission line is corrected in the ESPRESSO-NIRPS DRS while it is not in APERO DRS (Figure 2.14), which is expected because the line list of ESPRESSO-NIRPS is more than double the size of that of APERO, and this is one such line that is missing from APERO. One other thing to note, on the edge of order 55 (Figure B.6) there is a strong emission line that is left uncorrected. This is done on purpose because the location of the line prevents the interpolation algorithm employed in the correction to locate the entire wavelength region (Table A.2) and hence correct for the line. When this line appears in the next order 56, it is corrected without issues. When matched with the APERO DRS, this line on the edge of the order 55 seems to be better corrected on APERO, however no such differences occur in order 56.

2.10.3 TOI-4666 (Faint, HE mode)

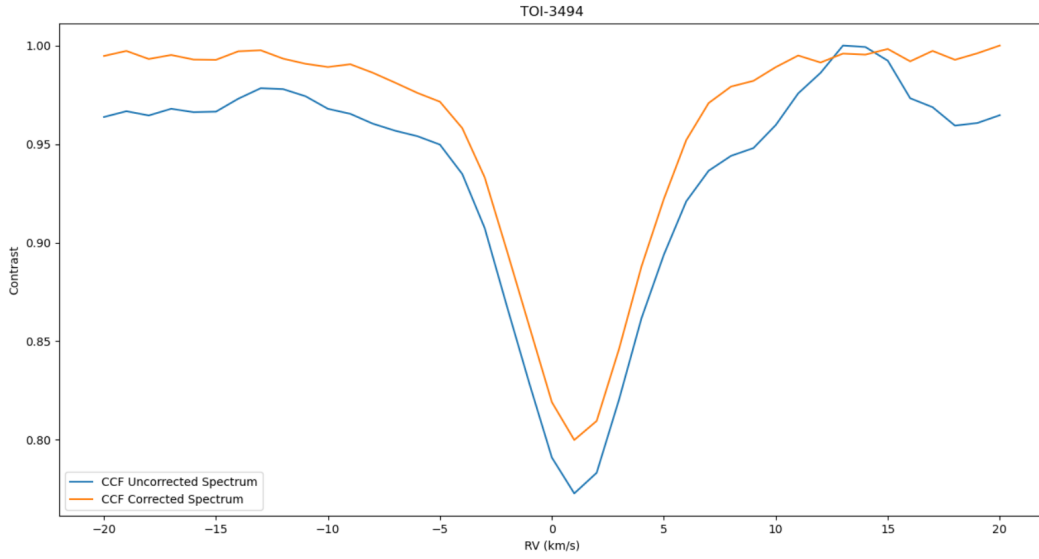
The final target to be thoroughly tested is TOI-4666, on the other extreme end of the NIRPS SNR range as Proxima as it is a faint M1 type star observed in HE mode. While the emission correction works, the SNR of this target is low enough that the correction does not work as effectively (Figure 2.15a) as it did for Proxima or TOI-3494, which is mainly noticeable in the CCF (Figure 2.15b) where the shape is not an extremely smooth downward facing gaussian. However, compared to what the CCF was like before the correction, it is a massive improvement.

The discrepancy between the python script and the DRS was similar to TOI-3494, of the level of 1-m/s while the RV error was in the tens of m/s, hence they were compatible. Here, the NIRPS M1 mask was used and possibly part of the reason the CCF looks skewed could be because the M1 mask is not that optimized and cleaned of telluric lines, however this is difficult to test for

2.10. VALIDATION OF EMISSION CORRECTION

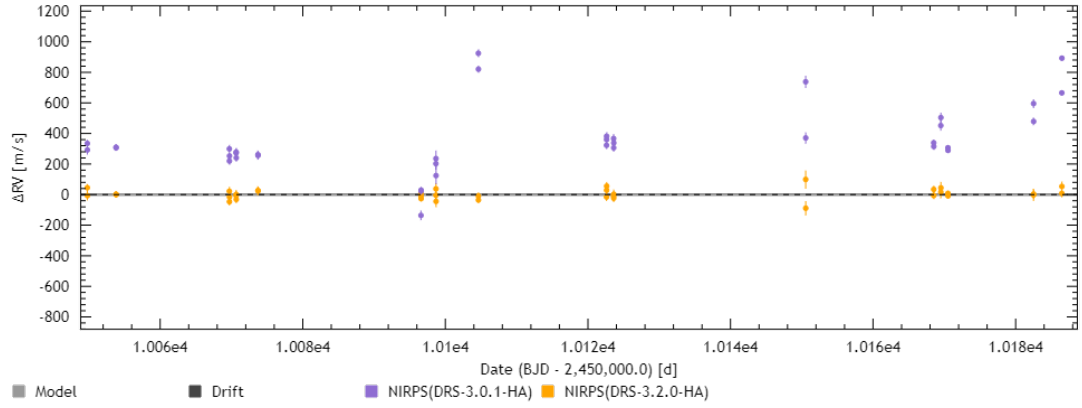


(a) Spectrum of HA mod TOI-3494 order 30 before and after emission correction.

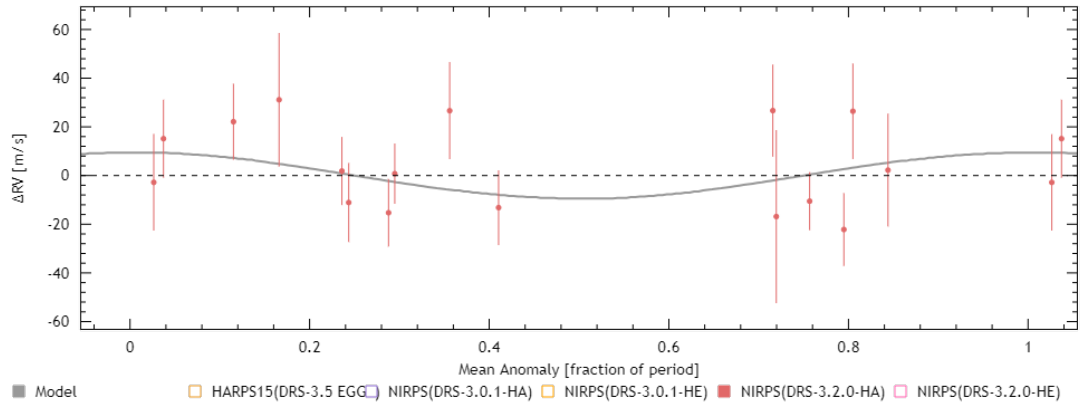


(b) CCF of HA mode TOI-3494 before and after emission correction. The bump around the 15-km/s mark is a clear sign of emission contamination. The RV scale here gives the relative RV to the systematic velocity input in the fits header of this frame.

Figure 2.12: Comparison before and after emission correction on HA mode target TOI-3494.



(a) Improvement in RV from DRS-3.0.1 to DRS-3.2.0 with DRS-3.0.1 showing clear scatter due to emission line contaminated CCFs.



(b) Timerseries of TOI-3494 from DRS-3.2.0 in agreement with orbital solution.

Figure 2.13: Comparison of RV timeseries before and after emission correction on HA mode target TOI-3494.

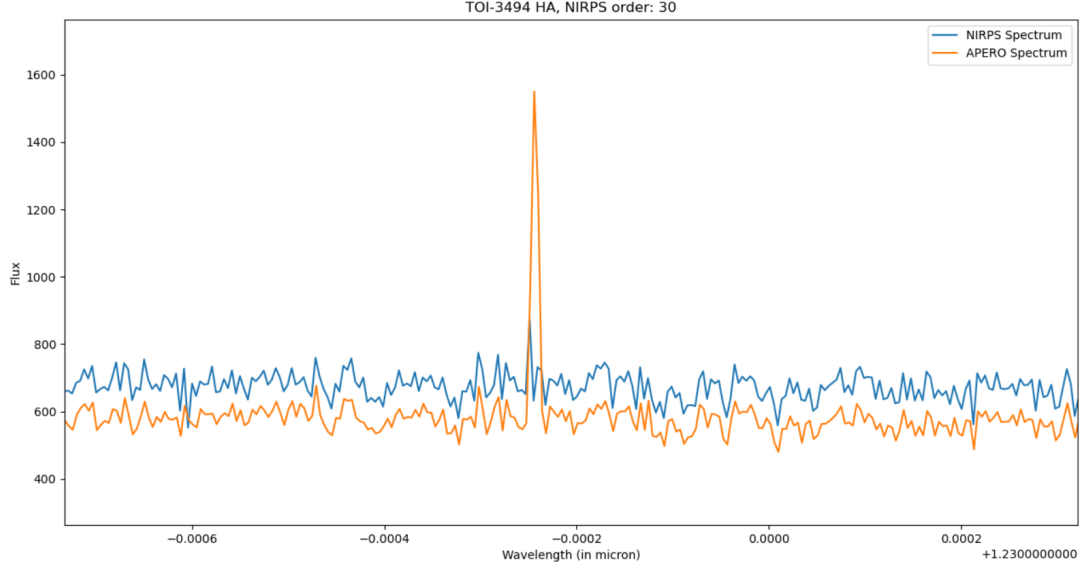


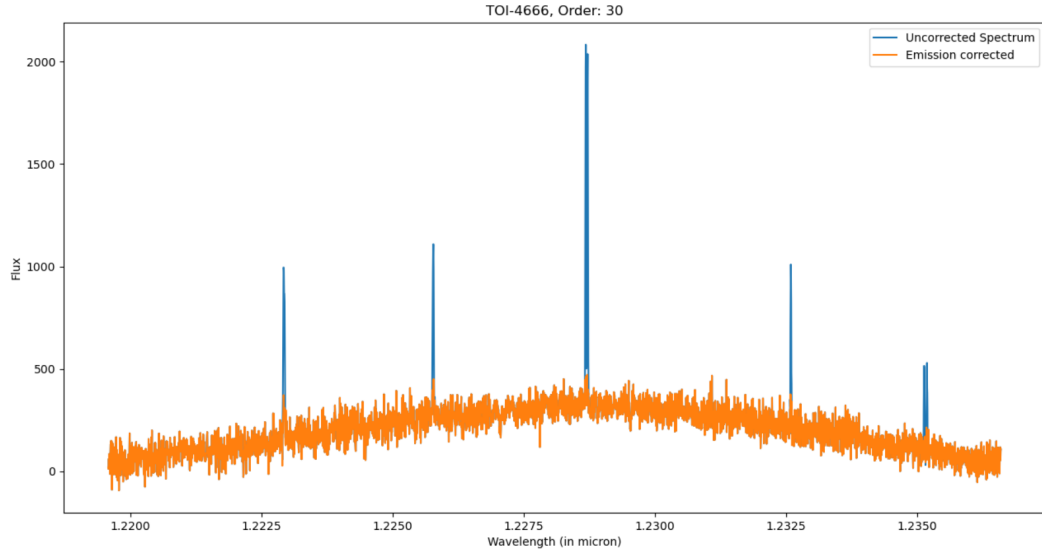
Figure 2.14: An emission line of HA mode TOI-3494 order 30 uncorrected in APERO but very well corrected in ESPRESSO-NIRPS DRS highlighting the difference in the line lists of the two pipelines.

such a low SNR target as the RV error is high enough to be of the same order of magnitude as the contamination induced by microtellurics. This is also one of the targets severely affected by the limitations (see Section 2.11) of this technique due to its low SNR and the clear emission residual seen on the CCF post-correction. Despite the setback, the planetary signal is clearly detected from NIRPS spectrum (Figure 2.16).

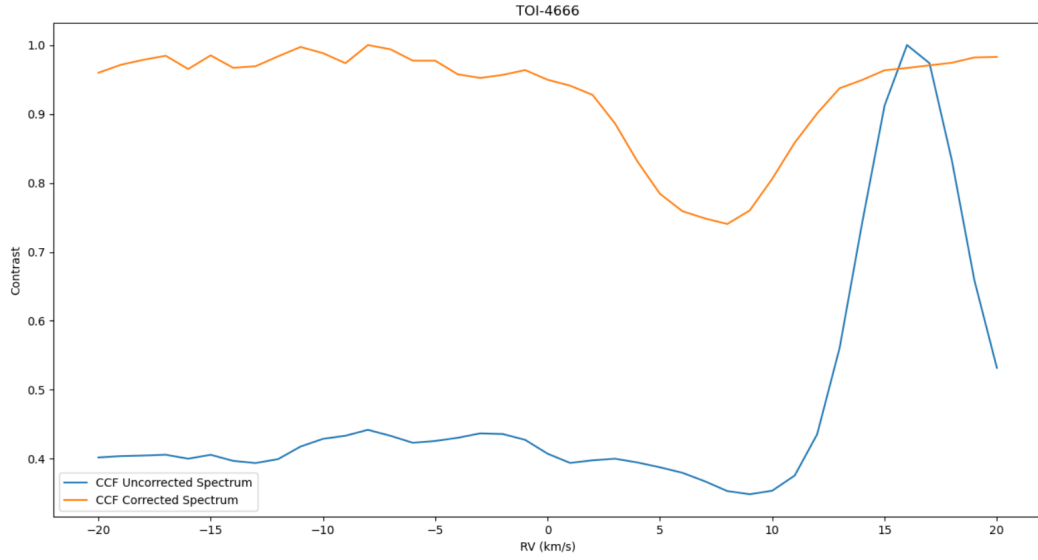
2.10.4 TRAPPIST-1 and TOI-756 (Faint, HE mode)

It is not enough to simply run the emission correction on two faint targets to conclude that it is working at the level of precision as desired, therefore additional tests were done on targets of similar levels of SNR, TRAPPIST-1 (faint, M7.7 type star) and TOI-756 (faint, M4 type star). From the inspection of the spectrum itself (Figures 2.17a and 2.17b) both targets have their emission features corrected adequately.

TRAPPIST-1 was also the first target on which the emission correction was tested during the developmental phase of the algorithm due to it being faint and the large number of times it was observed by NIRPS. Figure 2.18 compares CCFs developed using a custom mask where instead of stellar lines, the mask was composed of all the wavelengths of the emission lines. This is a good technique to test if these features were sufficiently corrected because if the correction worked perfectly, the resulting CCF should in theory be a flat line. Any contrast observed in the CCF post-correction acts as a measure (in percentage) of how well the correction works. Also, it helped in testing the various ratios used to correct the emission lines. When the per-line correction was used to reduce the effect of all the relatively stronger lines, the faint ones went uncorrected

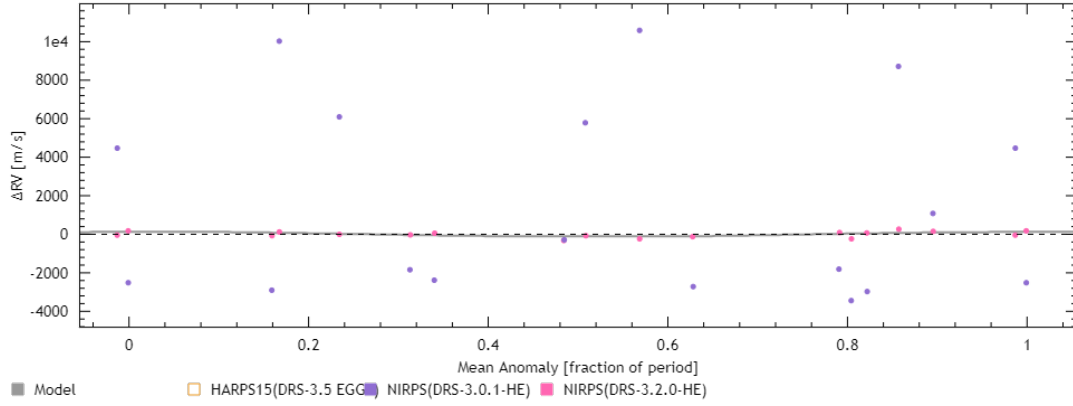


(a) Spectrum of HE mode TOI-4666 order 30 before and after emission correction.

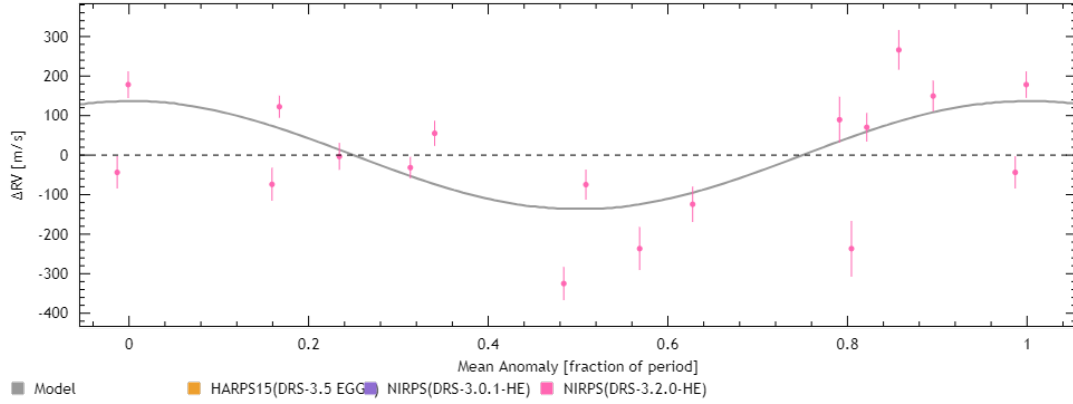


(b) CCF of HE mode TOI-4666 before and after emission correction. There is still some residual emission contamination in the CCF post-correction as the CCF is slightly asymmetrical.

Figure 2.15: Comparison before and after emission correction on HE mode target TOI-4666.

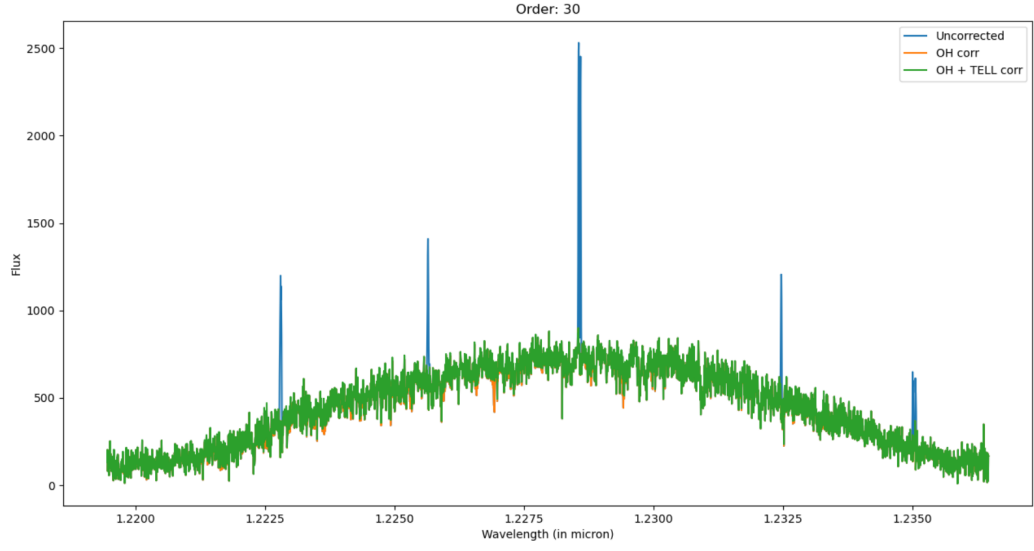


(a) Improvement in RV from DRS-3.0.1 to DRS-3.2.0 with DRS-3.0.1 showing clear scatter due to emission line contaminated CCFs.

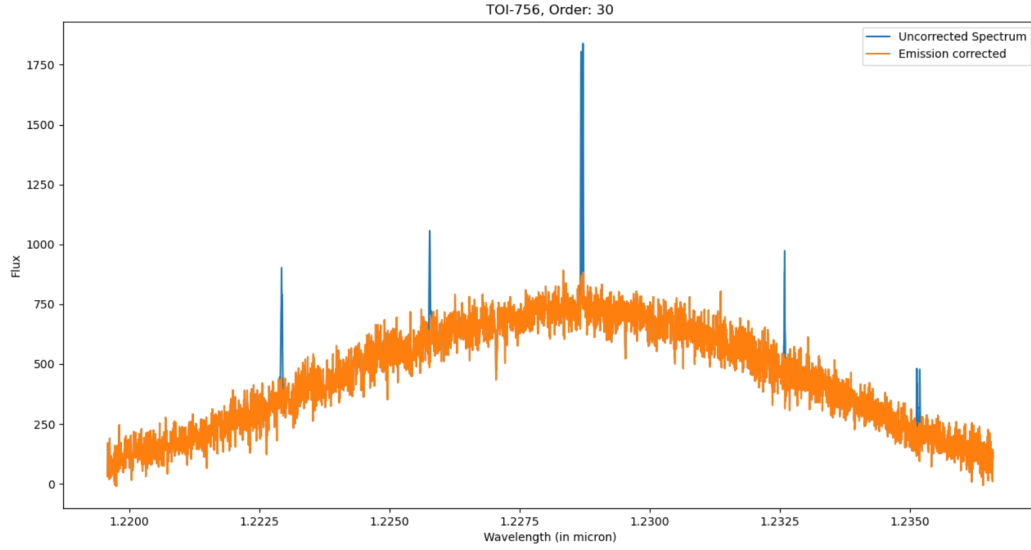


(b) Timseries of TOI-4666 from DRS-3.2.0 in agreement with orbital solution.

Figure 2.16: Comparison of RV timeseries before and after emission correction on HE mode target TOI-4666.



(a) Spectrum of HE mode TRAPPIST-1 order 30 before and after emission correction.



(b) Spectrum of HE mode TOI-756 order 30 before and after emission correction.

Figure 2.17: Comparison before and after emission correction on HE mode target TOI-4666.

and hence resulted in a contrast residual of approximately 10%. The common ratio and wave-range ratios also leave some residual when used by themselves. However, when per-line ratio was used in conjunction with wave-range or common ratio to correct for the lines that previously went uncorrected, the residual reached levels of $\sim 5\%$ for some targets, which is a significant improvement needed to produce good quality high-resolution spectra.

2.11 Limitations of the Technique

There is no doubt that the emission correction implemented in the ESPRESSO-NIRPS DRS pipeline has drastically improved the quality of spectra of faint stars in particular, producing better and accurate radial velocities (evident from the reduction in scatter of RVs of TOI-4666). As an automated process, there are bound to be scenarios where it does not function to its full capacity for a multitude of possible reasons despite our thorough testing. Listed below are some limitations and possible improvements that can be made to the current version of the emission correction implemented in ESPRESSO-NIRPS DRS-3.2.0.

2.11.1 Alternative Ratio for scaling the Master Spectrum

Previous studies (Rousselot et al., 2000) have concluded that the NIR emission of OH and O2 is the result of electron transitions between the different vibrational bands of these molecules. As such, the amplitudes of lines produced in the same transition scale similarly. The aforementioned study contains a detailed, theoretical prediction of the location and amplitudes of the emission lines, split according to the vibrational transitions that produce them. Using rough estimates, we were able to split the full NIRPS spectral range into wavelength regions according to Table A.1. This allowed for the usage of a different scaling ratio (wave-range ratio) to use instead of the common ratio for emission lines that cannot be corrected using the per-line ratio.

The variation of this ratio as a function of the vibrational band is depicted in Figure 2.19 for two different frames of TRAPPIST-1. From our analysis there appears to be no clear pattern in the wave-range ratios based on the vibrational bands. Regardless of the slight variations, if averaged across all vibrational bands, the wave-range ratio approximately equals the common ratio. The sharp increase in the wave-range ratio in the vibrational band number 6 can be explained by the 'break' in the NIRPS spectrum around order 42 and 43. As such, the neighbouring orders do not have the best signal to noise ratio and hence the values of the calculated ratios are inaccurate. It is for this reason we decided to drop the usage of wave-range ratio for the final version of the correction and only use the per-line and common ratios.

However, scientifically it would make sense to use this wave-range ratio. As stated above, rough estimates were used to split the entire NIRPS wavelength range into zones corresponding to the vibrational mode transition. Perhaps a more robust method would be to instead cross-match the emission line list of NIRPS with the one listed in the Rousselot paper and then assign each line its respective vibrational mode transition. Then in theory, all the lines with the same transition should have the same wave-range ratio. When we tried this in the earlier stages of this project, there was a large discrepancy between the lines detected on NIRPS and the ones in the line atlas, which is another reason why we decided to not pursue this route of investigation further.

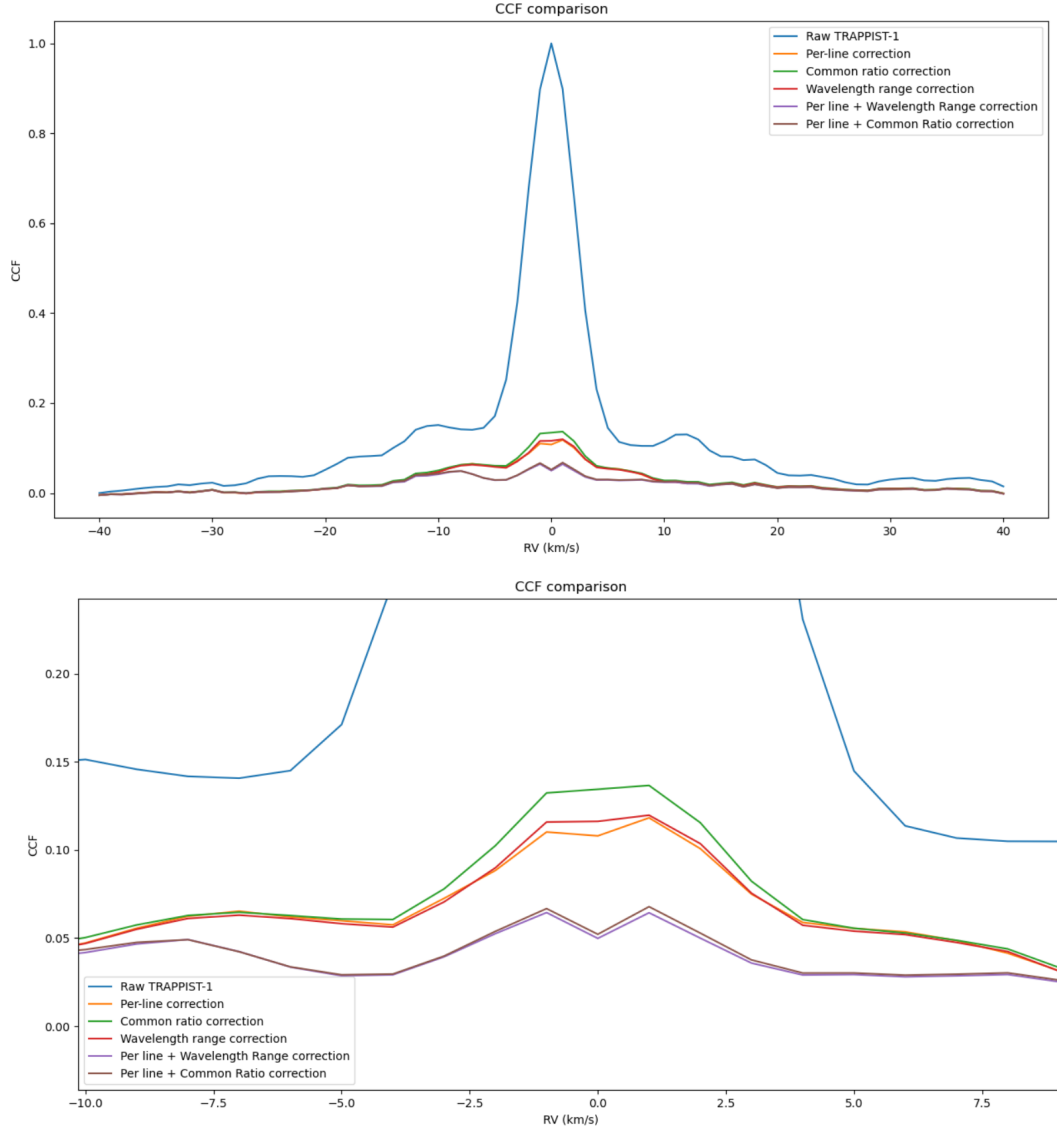


Figure 2.18: Comparing various ratios used to correct emission lines. This was useful in the early stages of the development of the algorithm. The per-line ratio needed to be paired with either the wave-range or common ratio to improve the correction to the 90-95% limit. Both cases give similar results and because the common ratio was easier to implement and justify from a scientific standpoint, the final version of the correction uses: per-line + common ratio.

2.11. LIMITATIONS OF THE TECHNIQUE

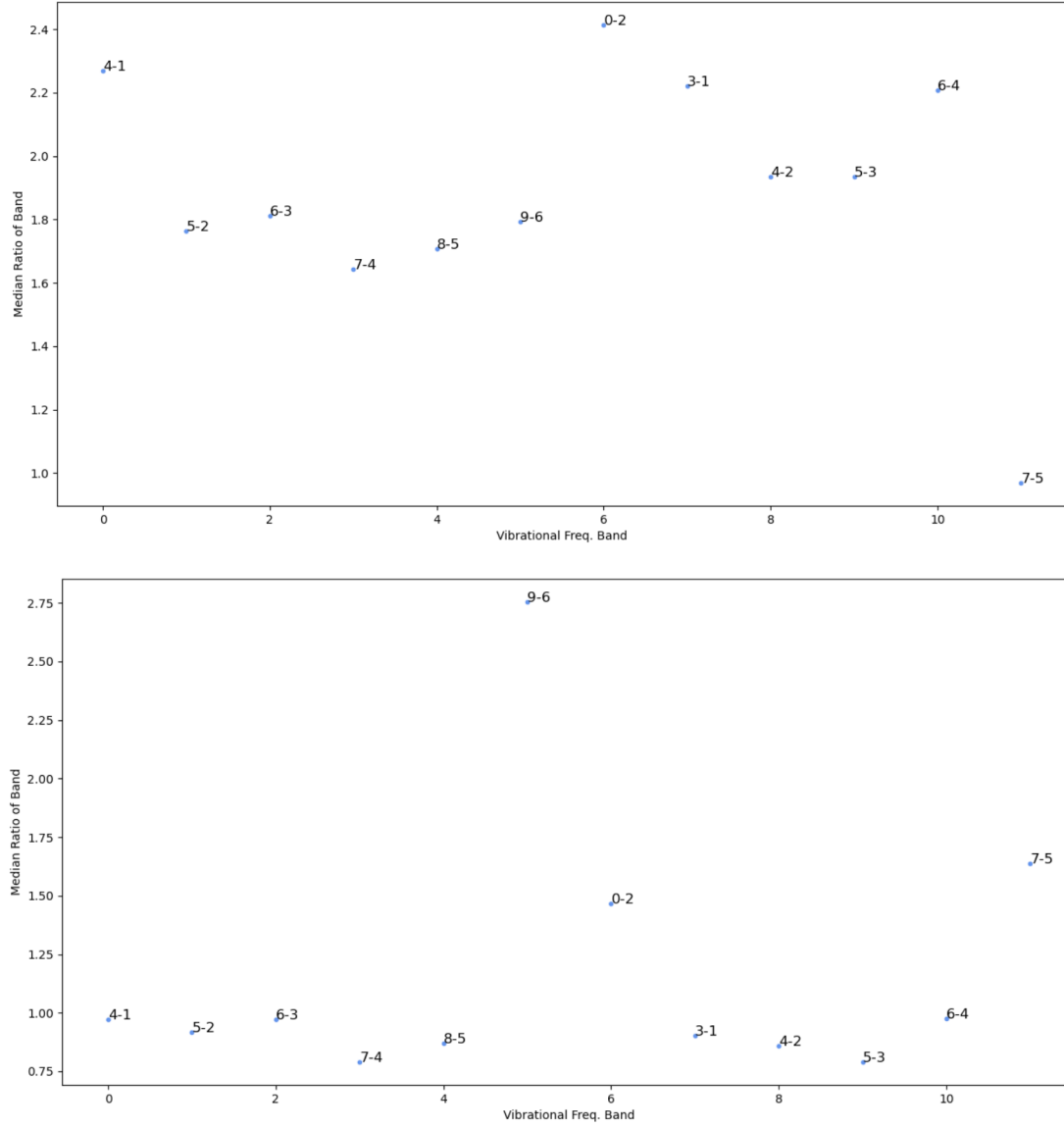


Figure 2.19: Two frames of TRAPPIST-1 comparing the wave-range ratios. Apart from the sharp increase in the ratio (transition 0-2) due to a gap in the NIRPS spectrum there is no notable pattern and this approach was discarded in favor of using a common ratio.

2.11.2 Under-Correction of Emission Lines at Larger Wavelengths

As mentioned previously, the emission correction leaves a 5-10% residual for very faint (low SNR) stars. Investigating this further, this residual is mainly present on the 'red' part of the NIRPS spectrum, comprised of the higher spectral order (50 - 70) as seen in Figure 2.20. It is surprising that there is always an under-correction and never an over-correction, which implies that there is more flux in Fiber A than expected, at least in the region of an emission line. There could be two possible reasons for this. First, Fiber A and Fiber B do not observe the exact same part of the sky. They are separated by 37-arcseconds which according to studies (Dauphin et al., 2024) is large enough for the sky to vary. Another possibility could be a change in the relative throughput of Fibers A and B over time. In Figure 2.21 the median flux ratio between Fiber A and B of all the emission lines for each frame used to construct the master sky spectrum is compared.

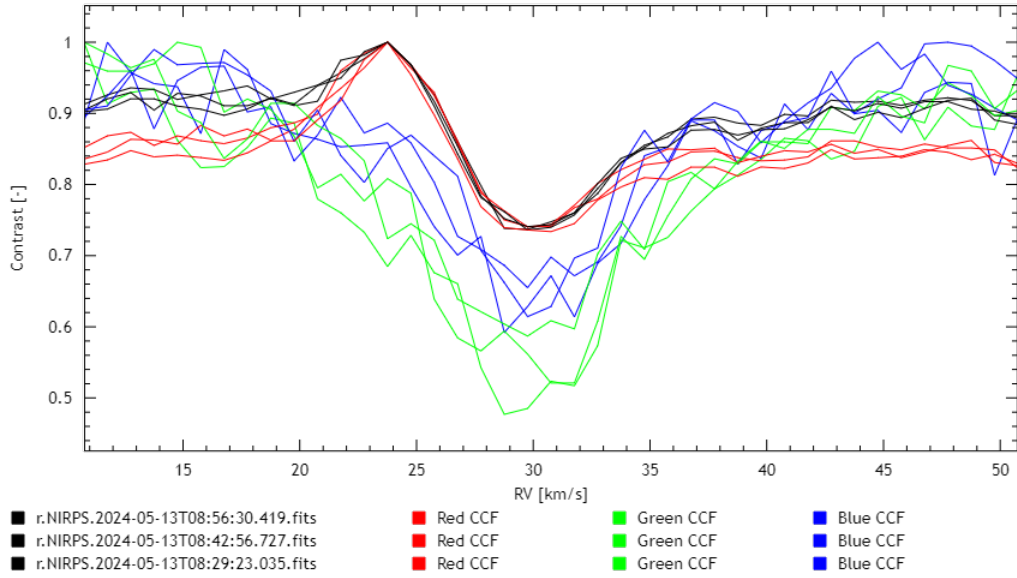


Figure 2.20: Clear under-correction of the emission lines in the red part of the NIRPS spectrum (orders 50-70) resulting in emission features still being slightly visible in the post-corrected CCF.

There is a clear scatter (approximately 5%) over one night and additional scatter over the course of several months. It is to be noted that during October and November 2023 there was an instrumental issue with the fiber stretchers of NIRPS, so it is not justified to use the sky-sky observations from within that time in the construction of the master spectrum. However, these frames comprise of 20% of the constituents of the master spectrum and removing them would result in a significant loss in SNR. To rectify this, we have now restarted sky-sky calibrations during technical nights to monitor the relative throughput between the two fibers and identify any trends that may or may not emerge, while also actively including these new frames in the construction of the master spectrum to increase SNR. Due to the limited time available to complete this project, the actual reason and solution to this under-correction is not yet explored.

2.11. LIMITATIONS OF THE TECHNIQUE

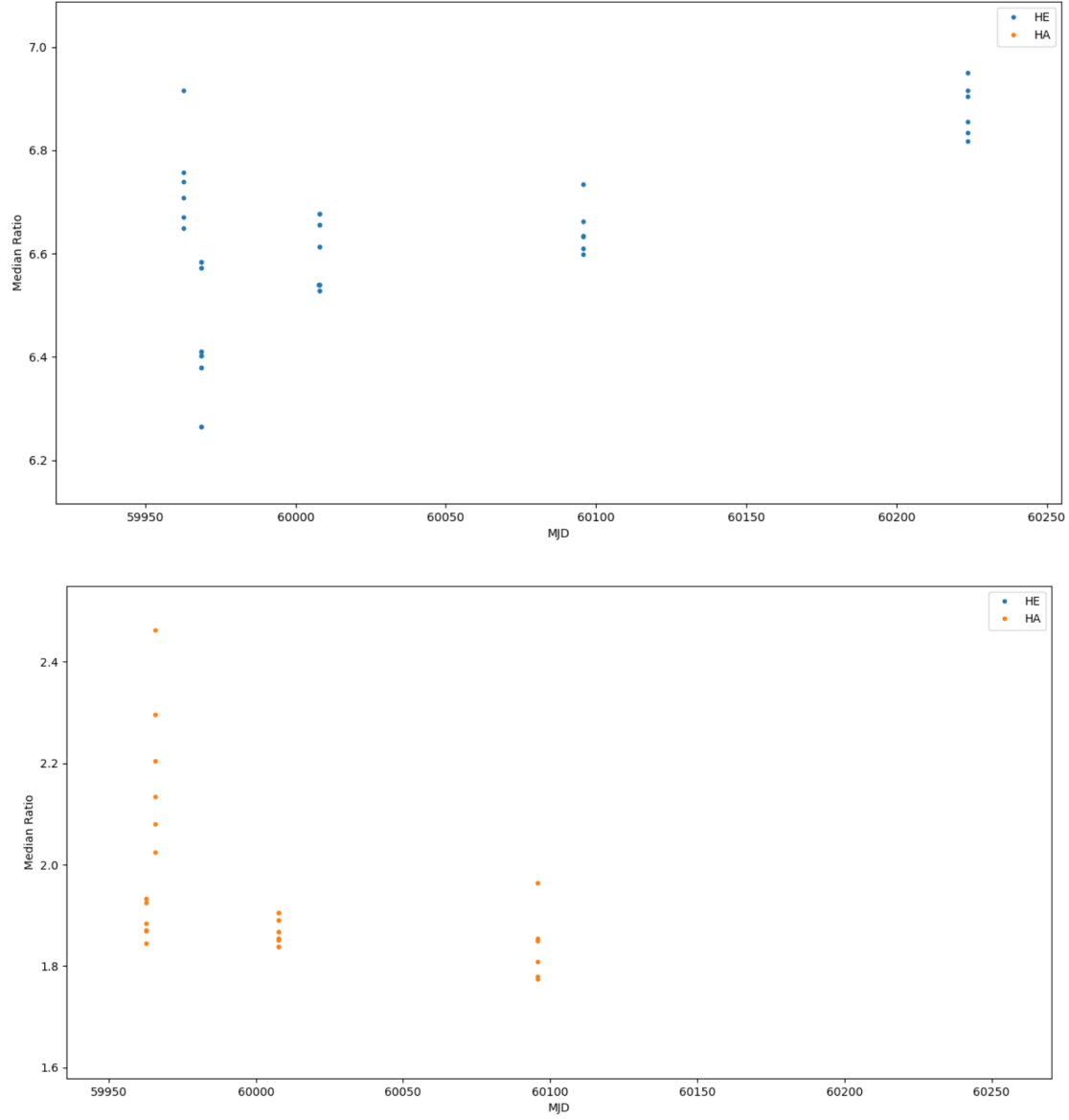


Figure 2.21: Plots showing the possible relative change in throughput of Fiber A and B for both HE and HA mode by comparing the flux ratios (as median ratio) of the emission features on all the sky-sky frames used in the construction of the master sky spectrum. For full list of frames see Table A.4

Chapter 3

Radial Velocities from Cross Correlation Function

3.1 Using CCFs in NIR

In the ESPRESSO-NIRPS DRS the radial velocities listed in the headers of the fits file come from the CCF data products (Table A.3). Due to the NIRPS spectrum having multiple spectral orders, a CCF is computed for each of them and then summed together to get the net result. This net CCF can then be normalized in order to view the contrast as a percentage. Masks derived from the following spectral type stars are available in the ESPRESSO-NIRPS DRS to compute the CCF: G2, G6, G8, K4, K6, M1 and M4.

There are many benefits and drawbacks of using the CCF to calculate RVs in the near infrared. On one hand, the shape of the CCF can give hints towards the possible existence of a spectroscopic binary and the FWHM can be linked to the stellar activity, but in the infrared the atmosphere is heavily contaminated by the atmospheric absorption and emission lines. Which implies that in the event these features are not corrected up to the desired precision, the CCF can become contaminated by telluric residuals and hence produce incorrect values for the FWHM, bisector and barycenter. These challenges prevent CCFs from reaching the 1-m/s RV precision for all the NIRPS targets.

3.1.1 Effect of Emission Correction on CCF

If a part of the spectrum of a target star has an emission line next to a stellar absorption line since the two peaks point in the opposite direction, the CCF contribution of this line may not be an inverted gaussian. Instead, if the contrast of the emission line is higher, the CCF contribution would resemble a right-side-up gaussian. This was the case for many of the fainter stars observed by NIRPS (see Figure 2.12b) before the implementation of the emission correction.

3.1.2 Effect of Telluric Correction on CCF

The effect of under-corrected, or in some cases uncorrected, telluric lines is a more subtle one on the CCF. Because the location of the stellar lines depend on the systemic velocity of the star and

the location of the tellurics depend on the Barycentric Earth Radial Velocity (BERV), the velocity of the Earth relative to the Sun, there is bound to be some overlap between the two types of lines. This overlap can result in the RV being correlated with BERV without showing a clear indication of this on the CCF. To test for this correlation, a timeseries (multiple RV observations over several nights) is required and since telescope time is limited, it is important that the tellurics and emission features are corrected for adequately to avoid this overlap in order to avoid bad RVs. Unfortunately, the correlation between RV and BERV is clearly evident in some of the bright targets (e.g. GJ643) so we experimented with a new method to optimize the CCF masks to minimize this effect induced by tellurics (see Section 3.2).

3.2 Modifying CCF Mask Weighting to Minimize Telluric Contamination

As mentioned previously, the CCF in the NIR is heavily influenced by Earth’s atmospheric contamination, despite correcting for the tellurics. A new method is proposed to reduce the effects of telluric lines when they coincide with stellar lines present in the CCF mask. The idea is to add a penalty to the ‘weight’ of the mask lines according to the contrast of close-by telluric line following the equation 3.2.1. Here, W' is the new weight, W is the default weight of the mask, m is a multiplicative factor, c is the penalty applied according to the contrast of the telluric line and n is a power factor for the penalty. There are three tune-able parameters in our algorithm: m , n , and the pixel range around the mask line over which we expect a strong telluric line to be present ($+/-p$). One can question the validity of this approach since it would require using a slightly altered CCF mask to compute the RV for each epoch, thus making it difficult to compare the RVs in a timeseries. However, we expect this so called ‘mask effect’ to introduce artifacts on the m/s level, whereas the effect of the tellurics can be attributed to being on the km/s level.

$$W' = W * m * c^n \quad (3.2.1)$$

Figure 3.1 shows the technique in action. The blue curve represents the telluric corrected spectrum of a star (GJ514 in this case) normalized to 1 only for visual representation, the orange curve is the telluric spectrum template, used to perform the telluric correction on the blue spectrum, the red line signifies the location of the CCF mask line and the green area is the selected pixel range. Since the peak of the telluric template overlaps with the stellar line, the chances of this line’s RV content being biased by the telluric is very high. To modify the mask line, the following steps are performed:

1. First the values of m , n , and p are defined. Since this is an experimental method at this stage, we do not have definite values for them. For the first testing, $m = 1$, $n = 1$ and $p = 1$ are adequate. Increasing the penalty may increase the RV precision, but it also increases the RV error so we had to be careful to balance the two.
2. Next, the algorithm picks a line from the CCF mask and records the wavelength. It then finds the pixel in the stellar spectrum with the wavelength closest to that of the mask line. Then depending on the value of p , a region is selected within which the telluric contamination is assumed to be maximum.

3.3. VALIDATION OF CCF MASK MODIFICATION ON RV TIMESERIES

3. Within the selected pixel region of $\pm p$, the minimum value of the contrast of the telluric spectrum is obtained. This becomes the parameter: c .
4. The value of c is then plugged into equation 3.2.1 and the new weight is calculated which is then used during the CCF computation. The entire process is repeated for every line in the CCF mask.

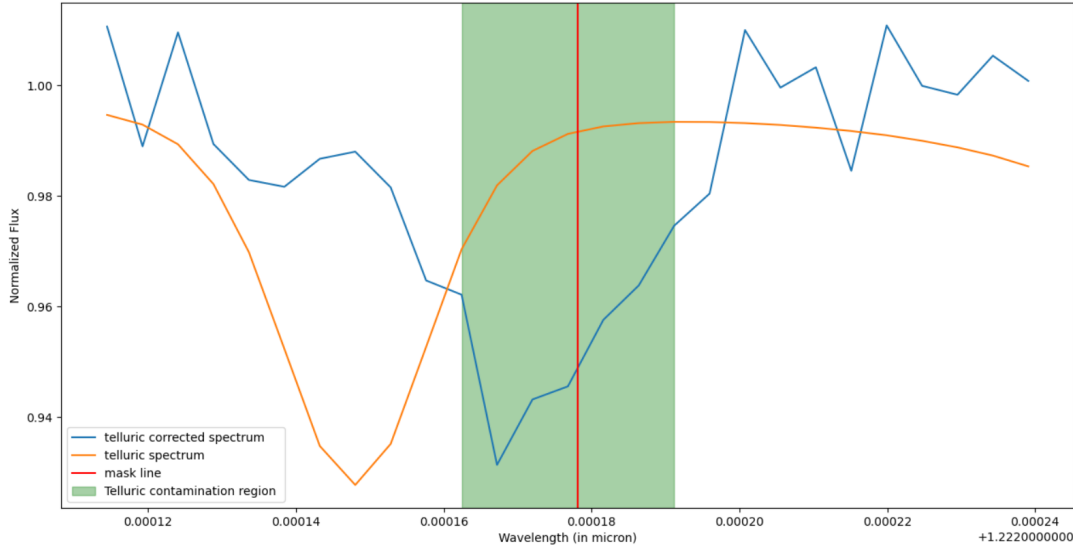


Figure 3.1: Visual representation of the technique of altering mask line weights to reduce telluric contamination on the CCF. The highest contrast (lowest value in flux) of the telluric spectrum located in the green region is used as the c parameter in equation 3.2.1.

To facilitate easier testing of this method, we decided to use a CCF computation code developed by Prof. Xavier Dumusque (hereby, unofficial CCF code) instead of making changes to the DRS pipeline. This code functions similarly to the CCF computation done by the DRS, however is one big difference. The unofficial code is somehow more sensitive to telluric contamination (see Figure 3.2) as the V_{tot} (RV - BERV) range of ± 10 -km/s is where microtellurics can affect the RV measurements and cause them to scatter the most. The exact cause of this is not known, but it is a difference regardless which is why, for consistency, we opted to regenerate all CCFs using the unofficial code and compare the results of the mask modification. It is clear that the official NIRPS CCF masks have not been completely cleaned of telluric features and it is imperative that this contamination is minimized.

3.3 Validation of CCF Mask Modification on RV Timeseries

It was noticed by members of the NIRPS Science Team that the CCF masks used in the official ESPRESSO-NIRPS DRS pipeline are not completely cleaned from telluric contamination, which

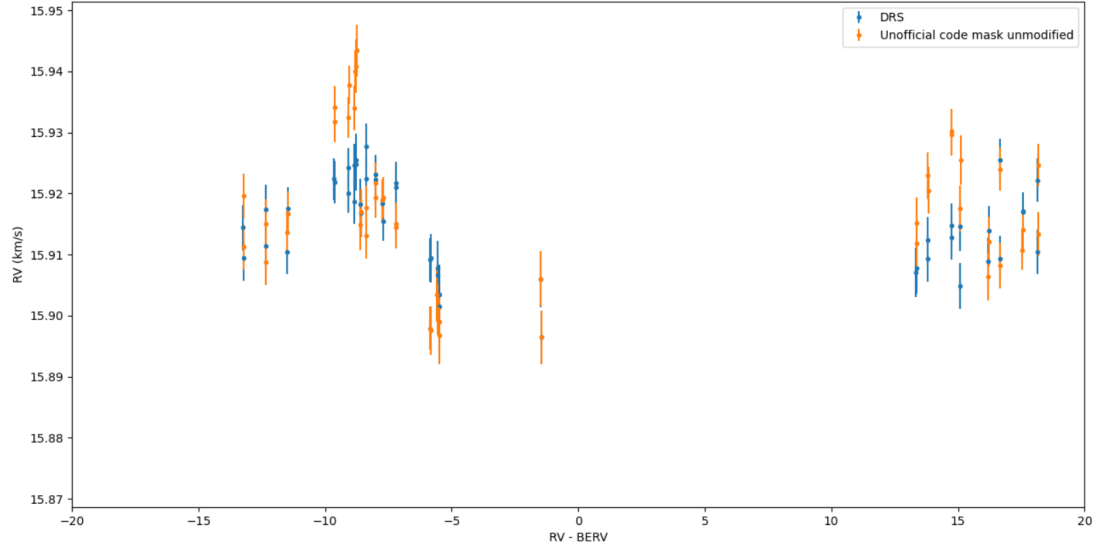


Figure 3.2: Plot of GJ643 in HE mode depicting the scatter in RV in the $|V_{tot}| \leq 10 - km/s$ range as a sign of telluric residuals affecting the CCF RVs despite the CCF masks being cleaned from tellurics. Also shows that the unofficial CCF code is more sensitive to telluric features compared to the DRS since the discrepancy lies in the V_{tot} (RV - BERV) range of ± 10 -km/s.

3.3. VALIDATION OF CCF MASK MODIFICATION ON RV TIMESERIES

is why we developed a method to do exactly that. This contamination was most significant in the V_{tot} range of ± 10 -km/s, where the effect of uncorrected microtelluric features was maximum. Unfortunately, the effect of the uncorrected or under-corrected tellurics was difficult to notice on the faint, low SNR targets as the photon noise level was higher. Locating a bright target star that has been extensively observed, with sizable measurements done in the required V_{tot} range was also no easy task. Finally, we settled on GJ514 (a bright telluric standard) and GJ643 (bright star part of the WP1 survey) that were the ideal candidates to test our method of modifying the CCF mask weighting and see if we are able to obtain better RVs.

3.3.1 GJ514

GJ514 is a bright M1 type star that is used as a telluric template, meaning it is a fast rotating star and hence has very broad and blended stellar absorption lines which leaves only the strong telluric absorption lines imprinted on the spectrum. This makes it an ideal candidate to be used as a reference in the telluric correction computation. GJ514 has been extensively observed because of this reason, however, there are not many frames that fall in the desired V_{tot} range, even though there appears to be some scatter, desired to test our method of altering the CCF mask weighting (Figure B.7) which is why we looked for another similar star.

3.3.2 GJ643

The perfect candidate to test the mask weighting method is GJ643. It is a part of the WP1 targets and is a bright M4 type star that satisfies the $V_{tot} \leq 10$ -km/s condition, as per (Figure 3.2). When $V_{tot} < -10$ -km/s the RV obtained from the DRS is fairly consistent, however soon after it crosses the threshold, we observe a scatter. This scatter is a clear sign of microtellurics going uncorrected and disrupting the CCF enough to create m/s level of scatter. The following set of Figures 3.3 show different scenarios with varying parameters in the formula that alters the contrast of the M4 mask lines that are located close to uncorrected telluric lines.

It is evident that even after pushing the contrast penalty to an extreme (Figure 3.4) such that the telluric contaminated lines are heavily down-weighted, the scatter still persists. There is some reduction in the scatter, but it is still within the error bars of the RV. One possible explanation could be that since the unofficial CCF code is more sensitive to tellurics, it is unable to incorporate the penalty correctly. Unfortunately, creating a new mask for every frame for the DRS to reduce the frame with is not an option, and due to the limited time frame of this project, this method cannot be explored further.

An alternative however emerged after inspecting the individual RVs of each spectral order for each frame of GJ643 (Figure 3.5a). Orders 0, 33 and 50 show extreme divergence in RV compared to the median obtained from other orders, which is expected because these are the orders that had the maximum number of telluric lines that had to be corrected for. Removing these orders completely from the final CCF does indeed reduce the scatter (Figure 3.5b) slightly and may act as a temporary solution to obtain precise RVs while the method of altering the mask contrast is tested further. However, the shift is still within the error bars so it is not a concrete conclusion. The most reliable method of fixing this contamination is to improve the automated telluric contamination algorithm.

CHAPTER 3. RADIAL VELOCITIES FROM CROSS CORRELATION FUNCTION

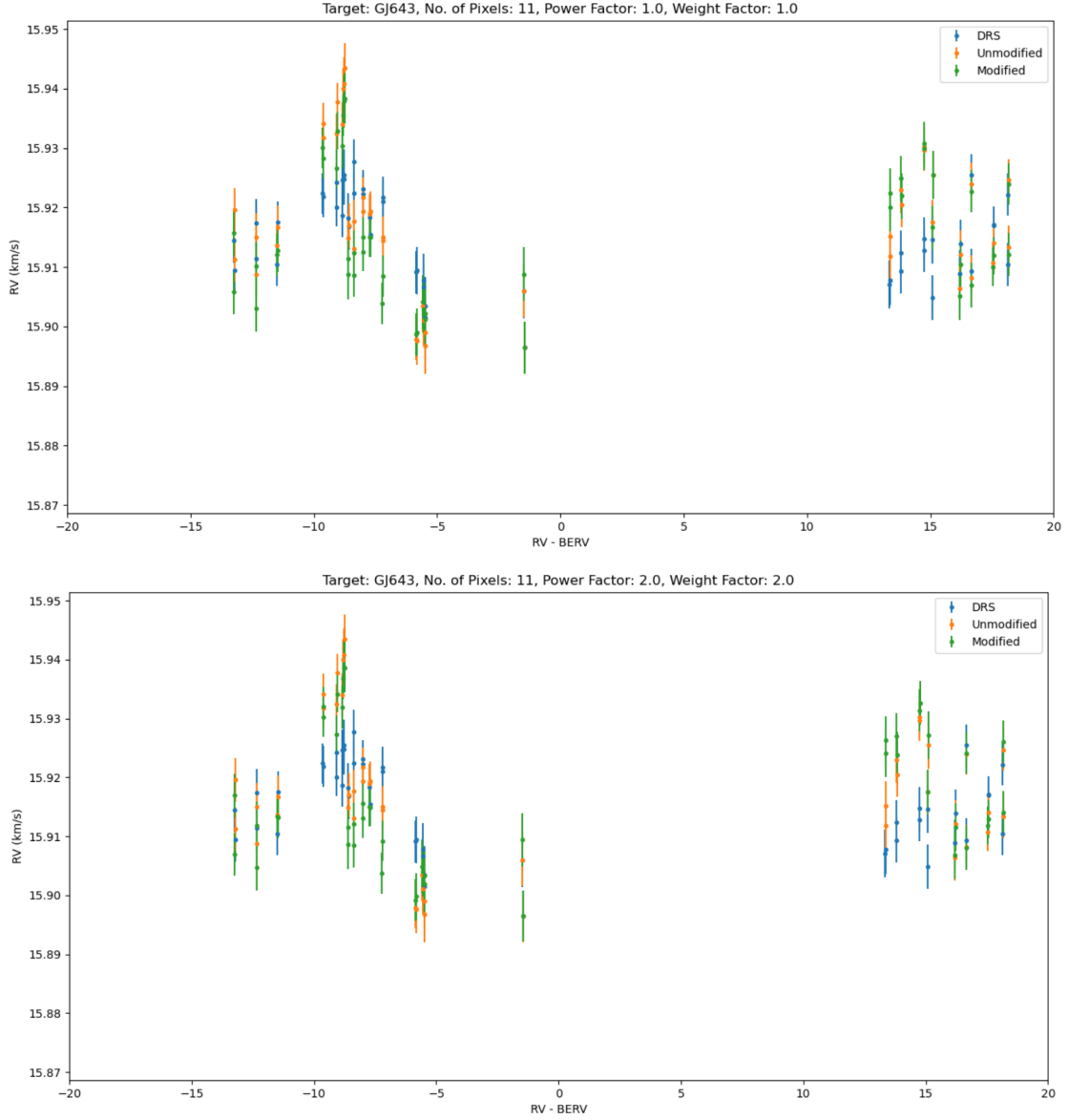


Figure 3.3: Variation of scatter in the RVs in the V_{tot} range of ± 10 -km/s for different parameters of the mask weight altering algorithm (more cases in Appendix). The 'Unmodified' and 'Modified' RVs correspond to those generated using the unofficial CCF code, which is too sensitive to the telluric features to effectively test this technique.

3.3. VALIDATION OF CCF MASK MODIFICATION ON RV TIMESERIES

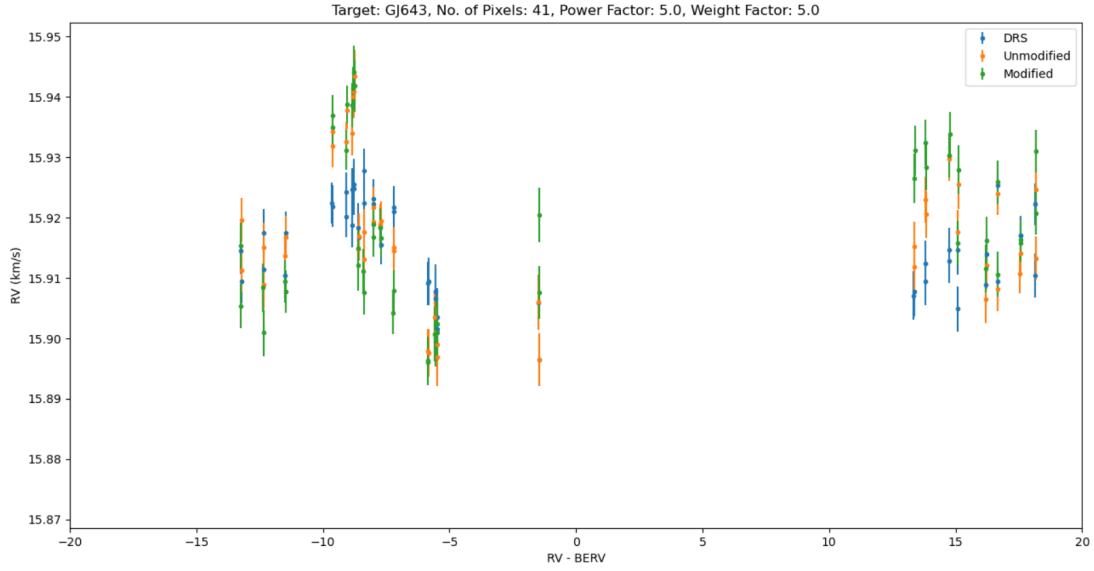
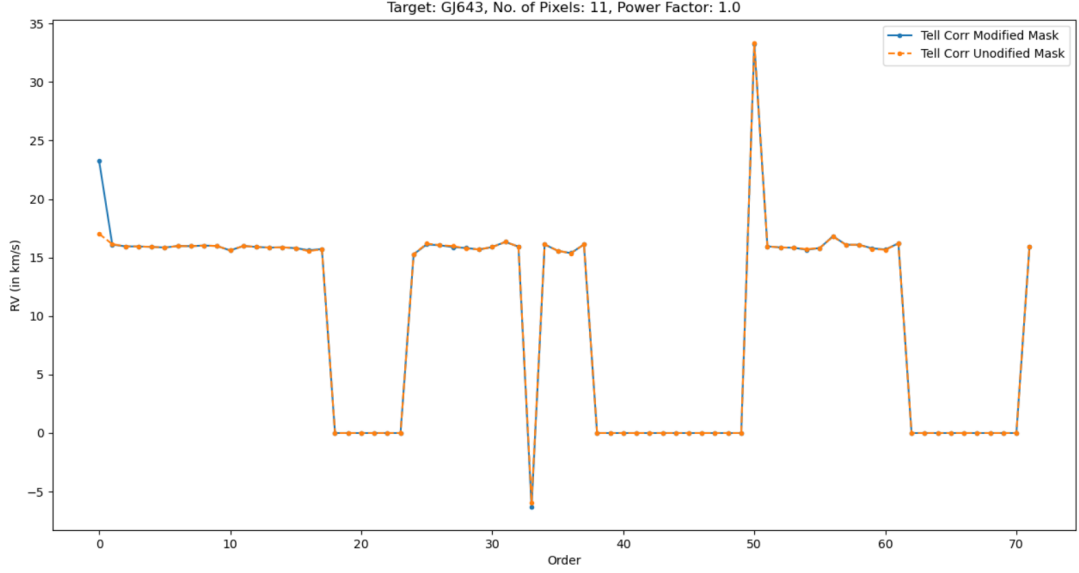
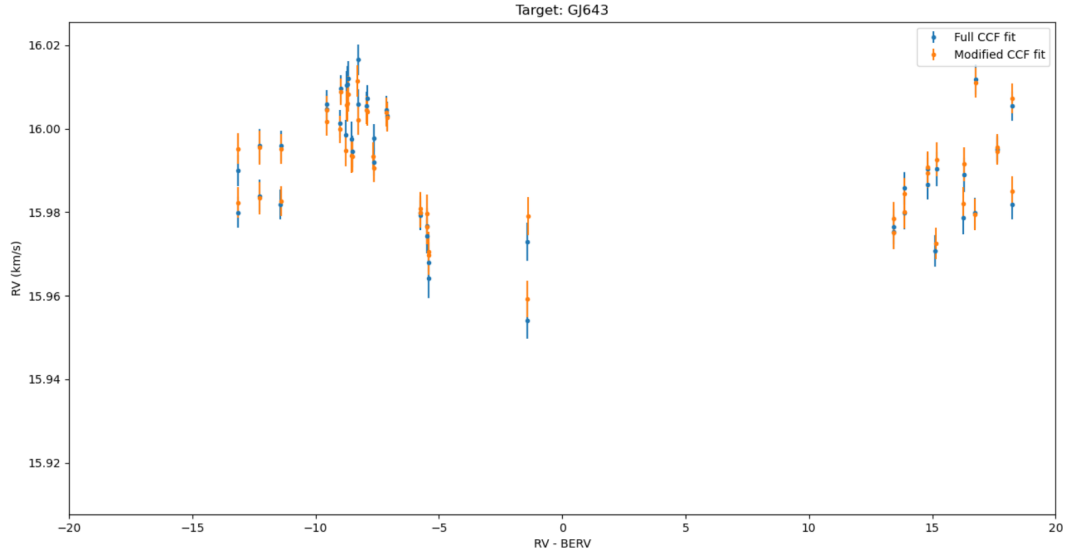


Figure 3.4: Plot of GJ643 in HE mode depicting the scatter in RV in the $|V_{tot}| \leq 10 - km/s$ range persisting despite pushing the mask weight altering parameters to the extreme. The 'Unmodified' and 'Modified' RVs correspond to those generated using the unofficial CCF code, which is too sensitive to the telluric features to effectively test this technique.

CHAPTER 3. RADIAL VELOCITIES FROM CROSS CORRELATION FUNCTION



(a) RV per order of a frame of GJ643 in HE mode. The orders with $RV = 0$ have no CCF mask lines present. Of the others, orders 0, 33 and 50 show the highest RV deviation due to telluric contamination even after modifying the mask weights.



(b) Comparison between the RVs from a 'full CCF' containing all orders and a 'modified CCF' with orders 0, 33 and 50 removed. The scatter due to tellurics seems to reduce but it is still within error bars.

Figure 3.5: Variation of scatter in the RVs in the V_{tot} range of ± 10 -km/s for different parameters of the mask weight altering algorithm (more cases in Appendix). The 'Unmodified' and 'Modified' RVs correspond to those generated using the unofficial CCF code, which is too sensitive to the telluric features to effectively test this technique.

Chapter 4

Conclusions and Perspective

4.1 Notable Results of the Emission Correction

A working version of the emission line correction was developed from scratch for the official ESPRESSO-NIRPS DRS to clean the NIR spectrum from the Earth's atmosphere's emission lines. This automated algorithm greatly improved the radial velocity measurements obtained from the Cross-Correlation Functions (CCFs) of faint stars. After testing it on a variety of targets, both faint and bright, and observed via both instrument modes (HA and HE) the contamination of the emission features was reduced by $\sim 90\text{-}95\%$. A line list was also generated with all the identified chemical species (mainly OH and O_2) that emit in the NIRPS wavelength regime.

Despite the positives, there are some limitations of the technique employed to do the emission correction. For very faint targets, the 5-10% residual appears to be quite significant and is visible as a bump in the CCF. This under-correction of emission lines always appears on the red part of the NIRPS spectrum (orders 50-70). Perhaps investigating further the wave-range ratios and using them instead of the common ratio would rectify this issue.

4.2 Notable Results from Modifying CCF Mask Weighting to Minimize Telluric Contamination

We also attempted to develop a method to reduce the impact of uncorrected or under-corrected telluric features on the RVs obtained from the CCF. The idea was to down-weight the CCF mask lines that overlapped with telluric features according to the strength of the telluric line, increasing slightly the RV error in return for more precise measurements. After thoroughly testing GJ643 in the V_{tot} (RV - BERV) range of $\pm 10\text{-km/s}$, the scatter in the RVs due to tellurics was not reduced by a significant amount. Instead, a greater impact was seen if spectral orders 0, 33 and 50 were removed from the CCF altogether. Another solution to minimize the telluric contamination would be to improve the current version of the telluric correction.

4.3 Future Works

4.3.1 Updates to Master Sky Spectrum

After analyzing the relative flux between the Fiber A and Fiber B of all the sky-sky observations used to construct the Master Sky-Sky spectrum, it was concluded that not all the frames were ideal. There was possibly some long-term trend in the relative throughput of the two fibers, some frames were also ill-suited because of an instrument error that persisted over two months. We have restarted sky-sky calibration observations to monitor any trends and add new frames to the Master Sky spectrum to increase the SNR and replace the problematic frames.

4.3.2 Implementing LBL Masks

Our colleagues in University of Montreal have developed new CCF masks compatible with the ESPRESSO-NIRPS DRS using data-products from the Line-By-Line RV analysis of some heavily observed stars matching our required spectral types. These masks have more lines (by a factor of 10) than the masks currently used and are much more customize in terms of altering the line weights and removing some lines altogether. The LBL masks would be prime candidates to test the algorithm that modifies the weight of the lines according to the telluric contamination and hopefully give more precise RVs through CCFs.

4.4 Perspective

With the newly implemented emission line correction in ESPRESSO-NIRPS DRS-3.2.0 the spectra and RV measurements have improved immensely, especially for faint stars. Not only does this benefit the NIRPS science team but also the entire astronomy community as the pipeline is publicly available. While one obstacle of the NIR has been tackled, the regime still continues to be a challenging domain for exoplanet detection, but with state-of-the-art instrument like NIRPS at the forefront, we are one step closer to the discovery of an Earth-like planet in the habitable zone of an M-type star.

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Appendix A

Tables

Vibrational Band	Wavelength Range (micron)	Orders (ESPRESSO -NIRPS)
4 - 1	0.9 - 1.07	0 - 4
5 - 2	1.07 - 1.123	5 - 18
6 - 3	1.123 - 1.195	19 - 26
7 - 4	1.195 - 1.275	27 - 34
8 - 5	1.275 - 1.36	35 - 41
9 - 6	1.36 - 1.43	42 - 44
2 - 0	1.43 - 1.47	45 - 47
3 - 1	1.47 - 1.565	48 - 53
4 - 2	1.565 - 1.62	54 - 56
5 - 3	1.62 - 1.725	57 - 62
6 - 4	1.725 - 1.83	63 - 66
7 - 5	1.83 - 1.94	67 - 70

Table A.1: Wavelength ranges corresponding to various vibrational bands of *OH* radical as a function of spectral order of NIRPS

Order	Wavelength Start (micron)	Region Size (micron)	Both Fibers	Species
0	0.98024	2.993e-05	0	Emission line wavelength(s) in region: [(0.98026, 'OH Line')] micron
2	0.99114	2.545e-05	0	Emission line wavelength(s) in region: [(0.99115, 'Unknown Species')] micron
2	0.99171	3.223e-05	0	Emission line wavelength(s) in region: [(0.99173, 'OH Line')] micron
2	0.99442	2.957e-05	0	Emission line wavelength(s) in region: [(0.99444, 'OH Line')] micron
2	0.9949	5.482e-05	0	Emission line wavelength(s) in region: [(0.99492, 'OH Line')] micron
2	0.99591	4.635e-05	1	Emission line wavelength(s) in region: [(0.99594, 'OH Line')] micron
3	1.00153	4.261e-05	1	Emission line wavelength(s) in region: [(1.00156, 'OH Line')] micron
4	1.00154	4.341e-05	0	Emission line wavelength(s) in region: [(1.00156, 'OH Line')] micron
4	1.00377	4.165e-05	0	Emission line wavelength(s) in region: [(1.00379, 'Unknown Species')] micron
4	1.00631	4.615e-05	0	Emission line wavelength(s) in region: [(1.00634, 'OH Line')] micron

Table A.2: Table containing the first few rows of the full emission line list used in DRS-3.2.0 of the ESPRESSO-NIRPS pipeline. The columns are as follows: Order - spectral order in the EPRESSO-NIRPS DRS where the emission line is detected on the master spectrum, Wavelength Start - the beginning of a line region in units of Angstrom, Region Size - the size, in Angstrom, within which the line is present, Species - the chemical species responsible for producing the emission line. The complete table can be found here: https://github.com/AvidaanSrivastava/UNIGE-Master-Thesis/blob/aa874e5d8a3027c41ab0e3438f1874e8a80d975e/Line_List.csv

Product Category	Description
CCF_A	CCF of uncorrected extracted spectrum of Fiber A
CCF_B	CCF of uncorrected extracted spectrum of Fiber B
CCF_SKYSUB_A	CCF of emission corrected spectrum of Fiber A
CCF_TELL_CORR_A	CCF of emission and telluric corrected spectrum of Fiber A
S2D_A	Extracted spectrum of each order of Fiber A, not flux calibrated, deblazed
S2D_B	Extracted spectrum of each order of Fiber B, not flux calibrated, deblazed
S2D_SKYSUB_A	Emission corrected spectrum of each order of Fiber A, not flux calibrated, deblazed
S2D_TELL_CORR_A	Emission and telluric corrected spectrum of each order of Fiber A, not flux calibrated, deblazed
S2D_BLAZE_A	Extracted spectrum of each order of Fiber A, not flux calibrated, with blaze
S2D_BLAZE_B	Extracted spectrum of each order of Fiber B, not flux calibrated, with blaze
S2D_BLAZE_SKYSUB_A	Emission corrected spectrum of each order of Fiber A, not flux calibrated, with blaze
S2D_BLAZE_TELL_CORR_A	Emission and telluric corrected spectrum of each order of Fiber A, not flux calibrated, with blaze
S2D_TELL_SPECTRUM_A	Telluric template generated from a bright star used in the telluric correction for the night

Table A.3: ESPRESSO-NIRPS DRS-3.2.0 data products relevant for this project

APPENDIX A. TABLES

row_id	name	night	frame_type	dpr_type	start_reduce_date	end_reduce_date	dfs_status	frame_group	tpl_wexp	tpl_expos	target	do_catg	qc	ins_mode
85426	NIRPS_2023-03-04T06:12:59.178 hrs	2023-03-03	SCIENCE-SKY	OBJECT-SKY	2023-05-16 10:36:23.029718	2023-05-16 10:38:15.034927	0	2023-03-04T06:12:29.21307	6	1	Night sky-sky- HA	{OBL_SKY}	1	HA
85433	NIRPS_2023-03-04T07:49:13.105 hrs	2023-03-03	SCIENCE-SKY	OBJECT-SKY	2023-05-14 18:34:04.591986	2023-05-14 18:35:32.196083	0	2023-03-04T07:18:08.21274	6	4	Night sky-sky HE	{OBL_SKY}	1	HE
85430	NIRPS_2023-03-04T07:04:04.465 hrs	2023-03-03	SCIENCE-SKY	OBJECT-SKY	2023-05-16 10:51:17.739209	2023-05-16 10:53:35.643879	0	2023-03-04T06:12:29.21271	6	6	Night sky-sky- HA	{OBL_SKY}	1	HA
88744	NIRPS_2023-01-18T04:45:07.552 hrs	2023-01-17	SCIENCE-SKY	OBJECT-SKY	2023-05-16 04:29:50.322228	2023-05-16 04:31:38.670486	0	2023-01-18T03:53:32.13470	6	6	Night sky-sky- HA	{OBL_SKY}	1	HA
106951	NIRPS_2023-05-31T06:09:17.983 hrs	2023-05-30	SCIENCE-SKY	OBJECT-SKY	2023-06-01 10:28:59.630811	2023-06-01 10:30:50.18324	0	2023-05-31T05:38:44.507	6	4	Night sky-sky HE	{OBL_SKY}	1	HE

Table A.4: List of NIRPS sky-sky frames used for the construction of the Master Sky spectrum. The full list can be found here: https://github.com/AvidaanSrivastava/UNICE-Master-Thesis/blob/22a9c2db7348338db3362cf5bfe6a10c01dd4e50/night_sky_sky_NIRPS.csv

Appendix B

Graphs

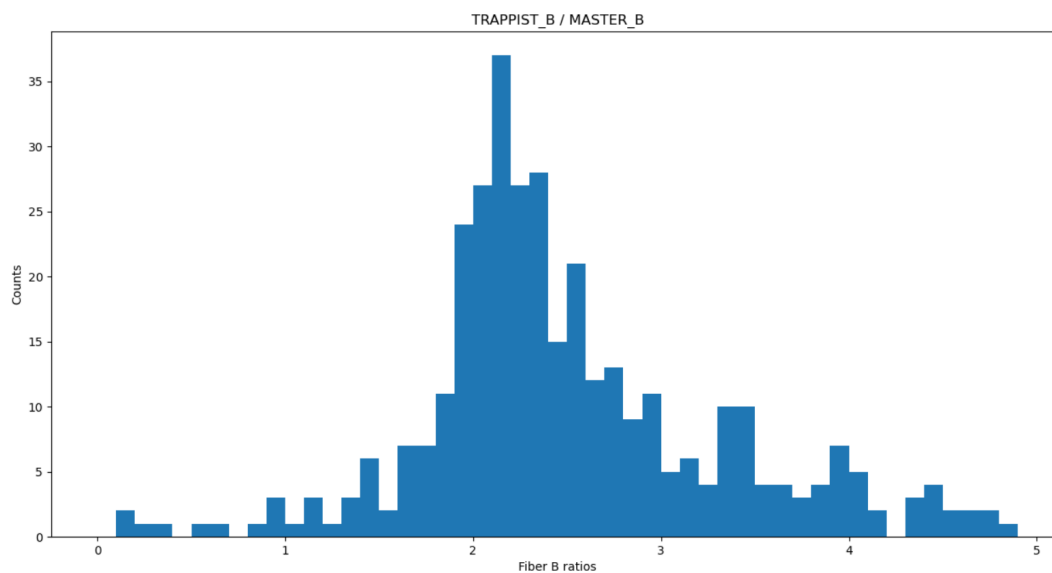
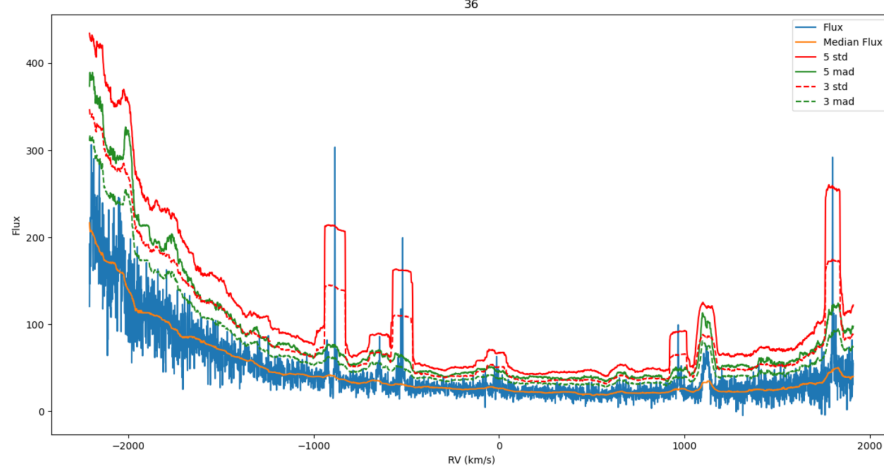
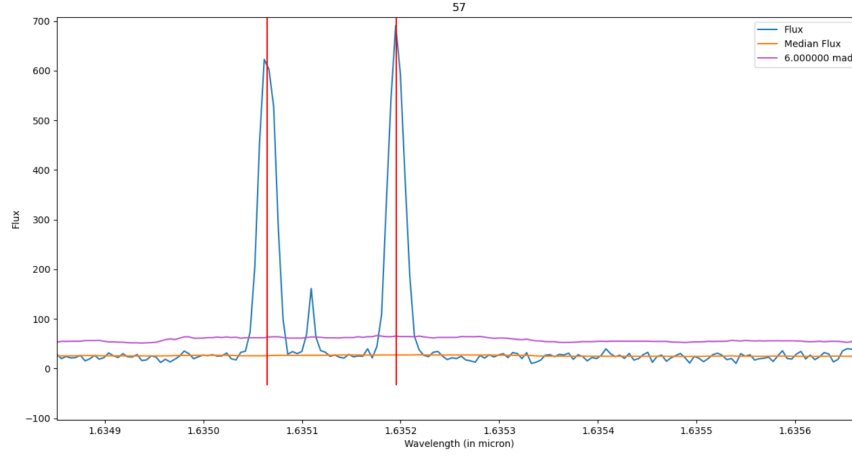


Figure B.1: Distribution of Fiber B per-line ratios of a frame of TRAPPIST-1 HE mode observation. Due to the wide spread, it is important to include the $0.33 \times$ common ratio (lower) and $1.67 \times$ common ratio (upper) bounds such that no absurd values of the ratio are being used for the correction. For this frame the common ratio = 2.4888 and the upper and lower bounds are 0.821 and 4.16 respectively.

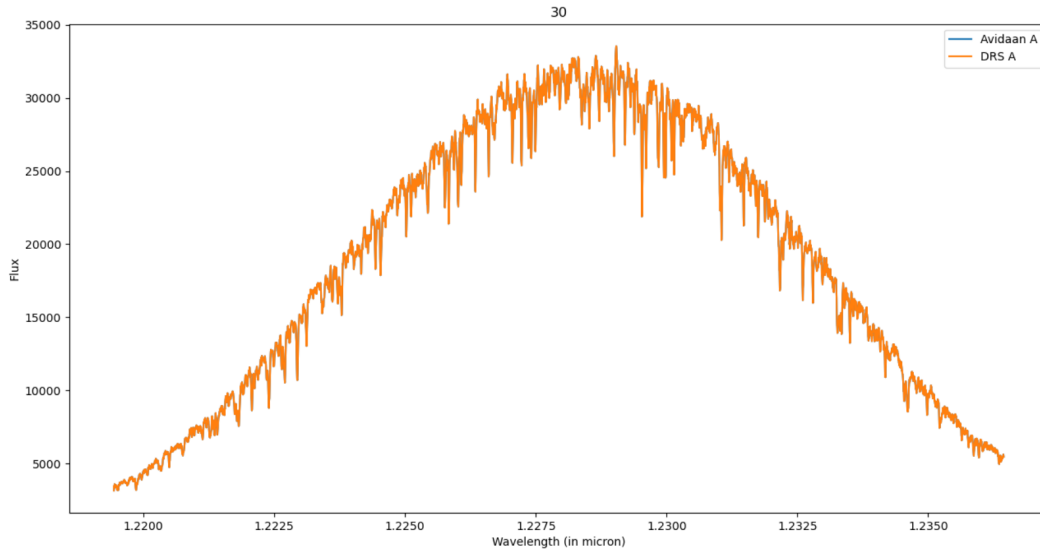


(a) Benefit of MAD (green) over STD (red). The value of STD jumps whenever there are emission features, while the rolling MAD stays at almost the same level above the continuum (Plot of deblazed master sky spectrum spectral order 36).

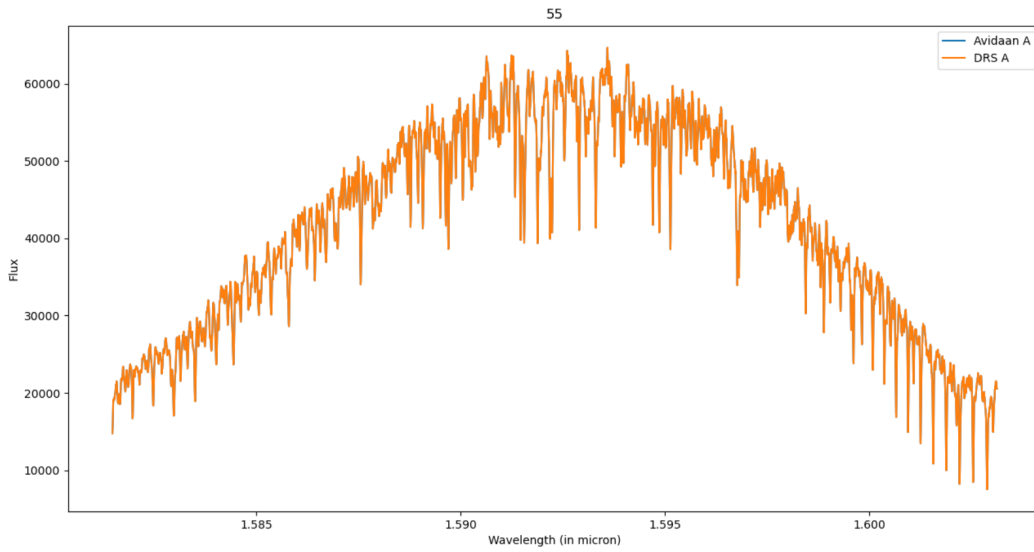


(b) $6 \times \text{MAD}$ limit detects best selection of lines from the testing done in this work. It is sufficiently above the continuum level but not too high such that the smaller emission features wouldn't get detected. It is however a free parameter and can be changed in the future if need be (Plot of deblazed master sky spectrum spectral order 37).

Figure B.2: Plots depicting why MAD was preferred over STD to identify emission lines and showcase the $6 \times \text{MAD}$ limit used for line selection.

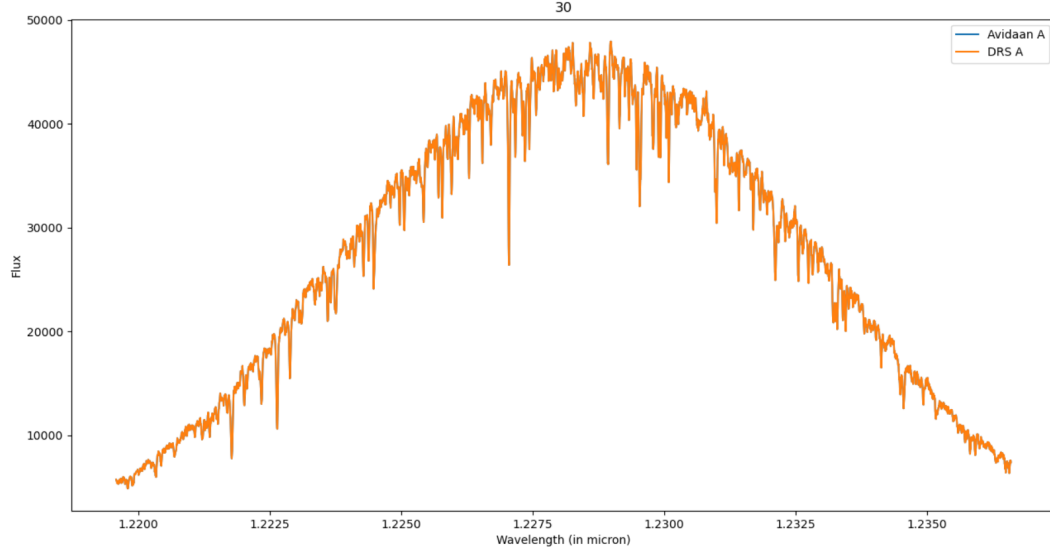


(a) Proxima HA mode spectral order 30

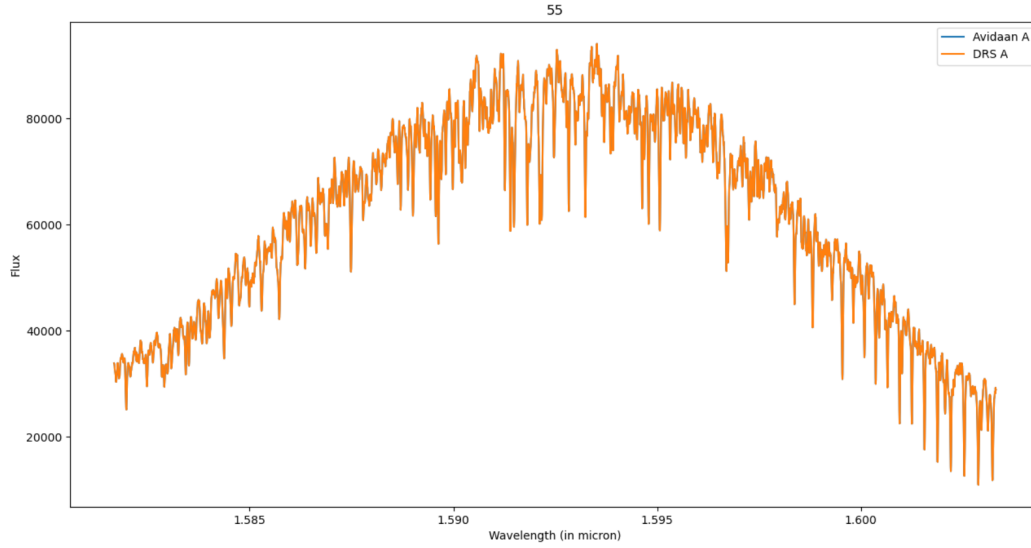


(b) Proxima HA mode spectral order 55

Figure B.3: Difference between the python script used to develop the emission correction algorithm and the output from its implementation in the official DRS pipeline.



(a) Proxima HE mode spectral order 30



(b) Proxima HE mode spectral order 55

Figure B.4: Difference between the python script used to develop the emission correction algorithm and the output from its implementation in the official DRS pipeline.

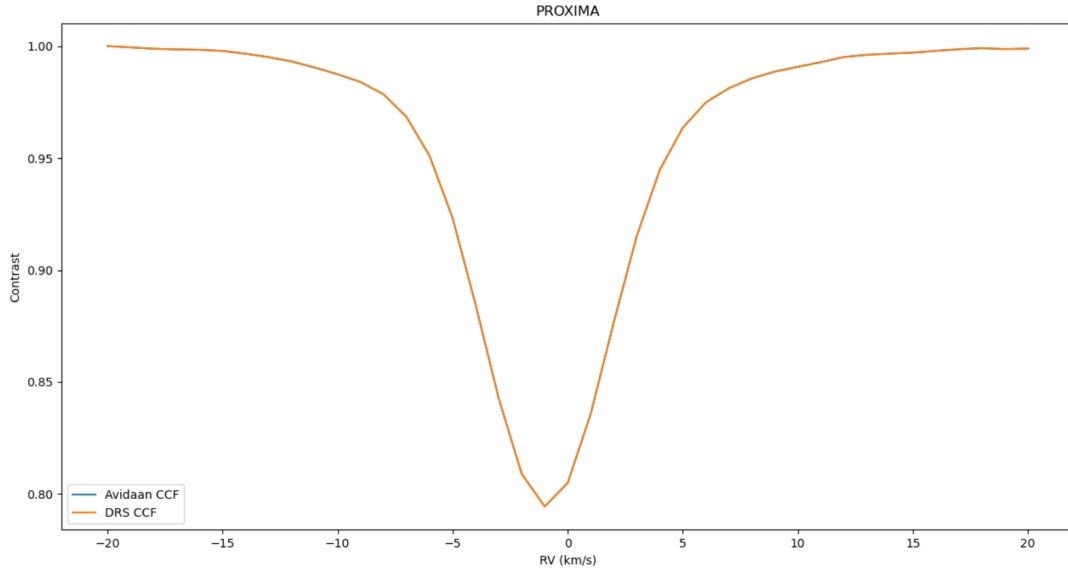


Figure B.5: CCFs generated from the emission corrected spectra computed using the python script and official DRS are in agreement up to 0.01-m/s level.

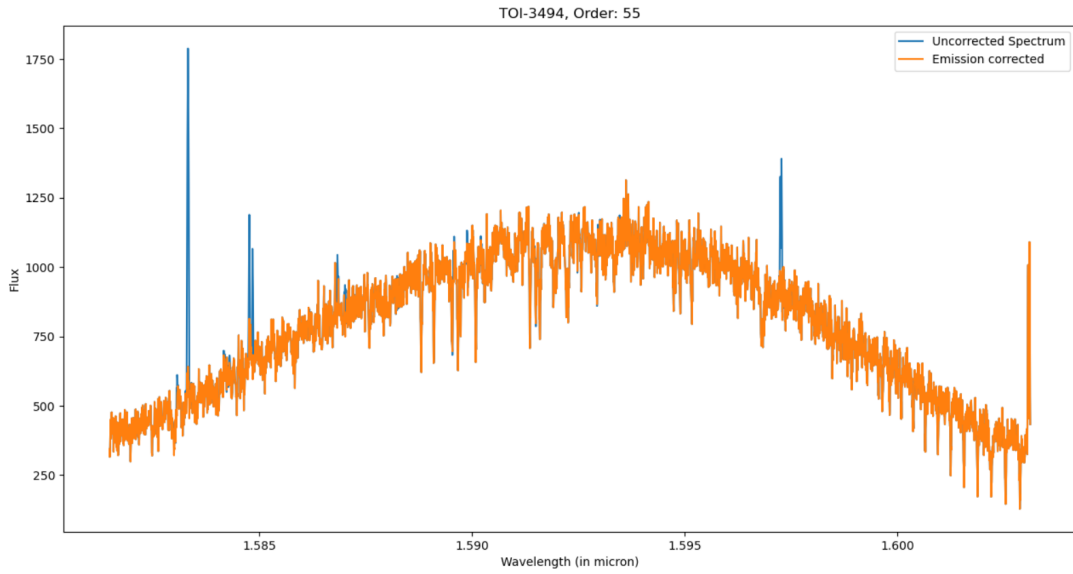


Figure B.6: Order 55 of TOI-3494 HA mode with an uncorrected emission line on the edge of the order. This was done on purpose because of constraints set in the DRS code.

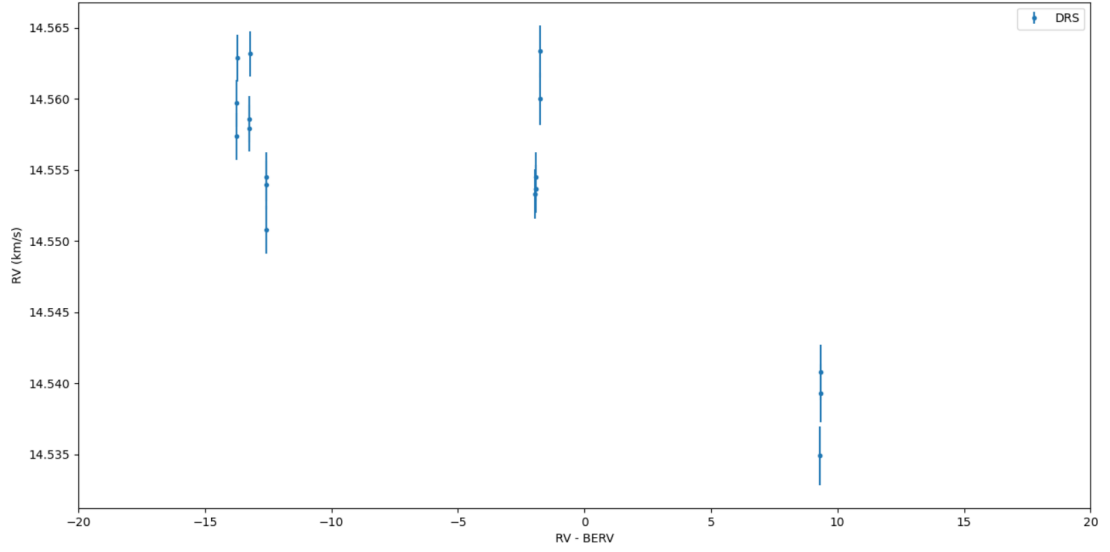
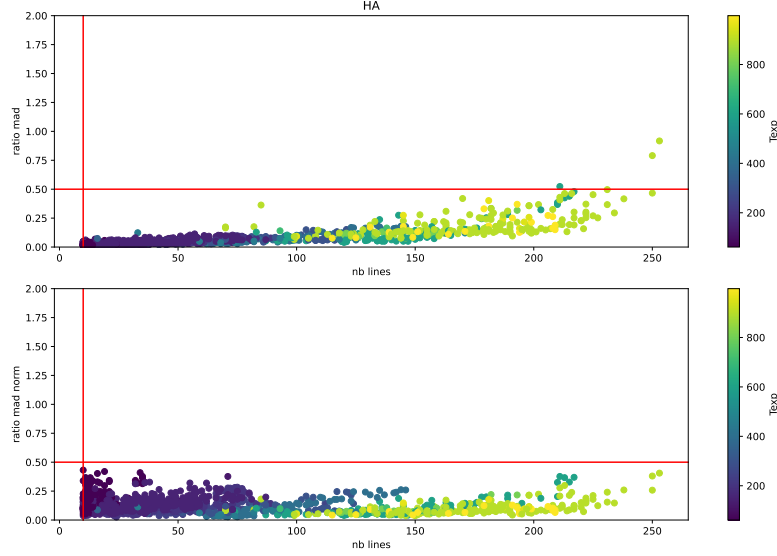
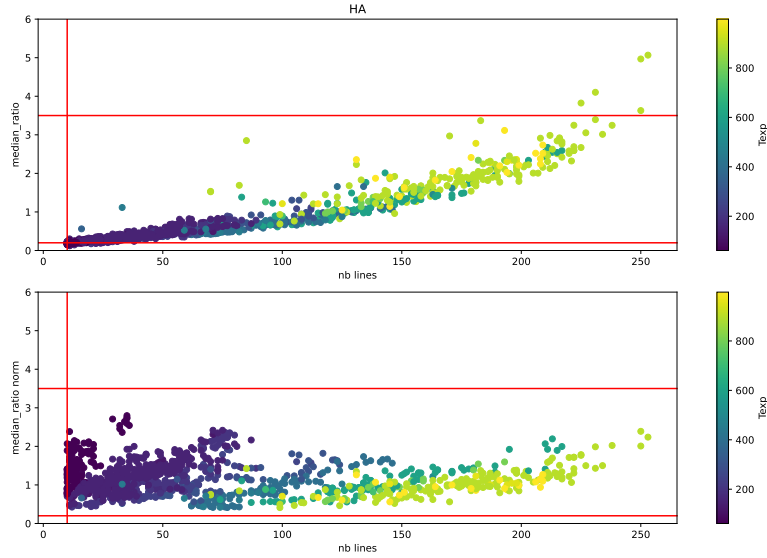


Figure B.7: Plot depicting the scatter in RV in the $|V_{tot}| \leq 10 - km/s$ range as a sign of possible telluric residuals affecting the CCF RVs despite the CCF masks being cleaned from tellurics of GJ514 in HE mode. There are not enough datapoints in this range to thoroughly test the technique.



(a) Normalized MAD Ratio QC bounds on HA mode spectra. The goal of the normalized MAD is to de-correlate exposure time and number of lines used to compute the common ratio to make all the frames comparable for QC checks.



(b) Normalized Median Ratio QC bounds on HA mode spectra. The goal of the normalized median ratio is to de-correlate exposure time and number of lines used to compute the common ratio to make all the frames comparable for QC checks. It seems there is still some correlation, but it is reduced enough to define the QC bounds.

Figure B.8: QC bounds on the Normalized MAD Ratio and Normalized Median ratio regarding the emission line correction. All data from 2022, 2023 and 2024 from NIRPS is represented on this figure.